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User Manual for Large Amplitude Oscillatory Shear (LAOS) Nonlinear Rheology Data Analysis Software (LAOS Analysis)

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Chapter 1: Software Overview

1.1 Development and Runtime Environment

This software is developed in Python and packaged as a standalone executable. It requires no pre-installation of Python or any third-party dependencies, allowing direct execution upon decompression.

1.2 Installation and Launch Instructions

The software is portable and requires no installation; it is ready for use immediately after extracting the downloaded archive. The extracted root directory contains the following components:

(1) LAOS_Analysis.exe: The main executable file.

(2) _internal folder: Contains the core dependency libraries and resource modules required for execution.

To launch the software, right-click LAOS_Analysis.exe and select "Run as administrator." (Note: To ensure proper execution, do not move LAOS_Analysis.exe out of the root directory, and avoid deleting or renaming the _internal folder, as doing so will prevent the program from launching.)

1.3 Design Rationale of the Software Logo

1.3.1 Overall Visual Tone

The logo primarily features a composed technological blue as its dominant color. Blue not only symbolizes the rigorous and rational essence of scientific research but also serves as a classic motif for fluids and oceans. This intuitively corresponds to the primary research subjects of the software—complex fluids and soft matter.

1.3.2 Analysis of Core Graphical Elements

The core graphic of the logo comprises an upper and a lower section, utilizing a minimalist visual language to precisely encapsulate the core physical processes and analytical methodologies of Large Amplitude Oscillatory Shear (LAOS) tests:

Central streamlined sinusoidal wave: The undulating flow pattern in the central region carries a dual metaphor. Firstly, the flowing wave morphology represents the research subject of "fluids," illustrating the continuous deformation and flow characteristics of materials under shear. Secondly, the wave strictly traces the periodic

trajectory of a sinusoidal function, directly signifying the foundation of LAOS testing—the oscillatory shear loading mode. The integrity and smoothness of the sinusoidal wave represent the precise application of periodic external forcing.

Bottom cascading mountain peaks: The "mountain peak" motif beneath the sinusoidal wave serves as a visual distillation of the nonlinear LAOS response in the frequency domain. Within the linear regime, the stress response consists solely of a single fundamental frequency; however, upon entering the large-amplitude nonlinear regime, the stress waveform undergoes distortion, exciting odd-order (e.g., 3rd, 5th, 7th) higher harmonic components in the Fourier transform spectrum. The bottom "peaks" serve as an abstract representation of these spectral peaks (i.e., variations in Fourier harmonics), where the undulating sequence of peaks symbolizes the superposition and evolution of higher harmonics in nonlinear viscoelastic behavior.

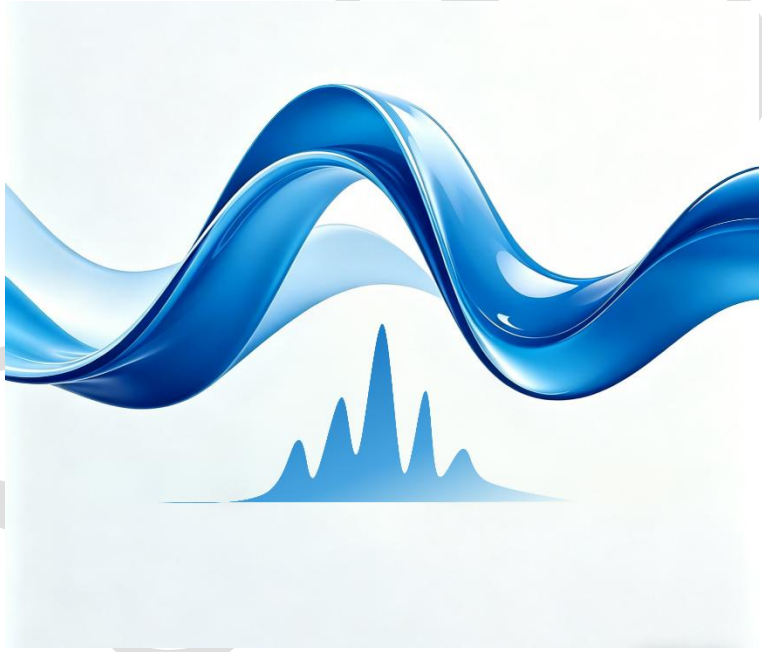


Figure 1. Logo of the LAOS Analysis software.

1.3.3 Integration of Overall Imagery

The top-down visual transition from the "sinusoidal wave" to the "cascading mountain peaks" establishes a logical progression from time-domain signals (periodic oscillations) to frequency-domain features (extraction of Fourier harmonics). The logo seamlessly integrates the dynamic aesthetics of fluids with the rigor of mathematics, succinctly encapsulating the core mission of the LAOS Analysis software: to accurately

extract frequency-domain parameters from complex time-domain oscillatory signals for the in-depth characterization of nonlinear rheological behaviors.

1.3.4 Intellectual Property Statement

This logo is an original visual identity designed for the LAOS Analysis software. All graphical elements and color palettes are abstractly derived from the software's inherent academic functions and underlying physical concepts. The design does not reference, adapt, or incorporate any names, trademarks, brand logos, or copyrighted visual elements belonging to any other legal entities, organizations, or individuals, thereby presenting no risk of intellectual property infringement.

1.4 Software Functionality

Large Amplitude Oscillatory Shear (LAOS) testing is a fundamental technique for characterizing the nonlinear viscoelastic behavior of complex fluids. However, the processing and cleaning of massive raw datasets, the variability and complexity among different LAOS analytical methods, and the construction of multidimensional visualizations often hinder the broader application of LAOS analysis across various fields. LAOS Analysis is a dedicated software designed specifically for LAOS data, aiming to enhance data processing efficiency and enable the precise extraction of various nonlinear rheological parameters from complex raw signals.

The software yields the following outputs: Lissajous curves; stress and strain normalization; dissipation ratio; Fourier transform; S and T values (Chebyshev polynomials); stress decomposition; Sequence of Physical Processes (SPP); apparent cage modulus; and contact stress decomposition.

1.5 Key Features

The core features of the software include:

(1) Comprehensive Data Preprocessing: Covers all preliminary steps required for LAOS analysis. It supports automatic raw data segmentation (for extracting steady-state cycles), noise reduction (filtering high-frequency noise and environmental interference), phase correction (eliminating phase drift caused by instrument inertia), shear rate calculation, and Fast Fourier Transform (FFT), providing a reliable foundation for subsequent nonlinear feature extraction.

(2) Broad Multi-Instrument Compatibility: Addresses the issue of heterogeneous data formats. It directly imports dynamic oscillatory data from mainstream rotational

rheometers (e.g., HAAKE, TA Instruments) and is compatible with standard formats (TXT, CSV) as well as proprietary manufacturer formats, eliminating the need for manual format conversion prior to analysis.

(3) Multidimensional Visualization: One-click generation of essential LAOS plots, including Lissajous curves, Fourier spectra, stress decomposition curves, and SPP Cole-Cole plots, facilitating intuitive representation of nonlinear characteristics.

(4) Multilingual Interface: Supports seamless switching between Chinese, English, and Japanese. This accommodates daily use by domestic researchers while facilitating international collaboration. The graphical interface also allows for direct export of publication-ready figures.

1.6 Application Domains

In conventional rheological testing, Small Amplitude Oscillatory Shear (SAOS) is commonly employed to probe the linear viscoelastic region (LVR) of materials. Within the LVR, the applied strain is minimal, inducing only reversible deformation of the internal microstructure; thus, the test results primarily reflect the undisturbed intrinsic structure of the material. However, during practical processing (e.g., extrusion, coating, spraying) and application (e.g., mastication, spreading, pumping), soft matter and complex fluids are typically subjected to large deformations. Their internal structures undergo irreversible destruction and reconstruction, thereby entering the nonlinear regime. Under these conditions, SAOS fails to faithfully reflect the structural evolution stress states of the materials. Large Amplitude Oscillatory Shear (LAOS) technology is specifically designed to investigate the nonlinear rheological behavior of materials, compensating for the limitations of conventional testing under large deformation conditions [1, 2].

By applying a sinusoidal oscillation with an amplitude well beyond the linear critical strain (γ_c), LAOS forces the material into the nonlinear viscoelastic region. In SAOS, the stress response is a sinusoidal wave containing only the fundamental frequency. In contrast, under LAOS, the stress waveform distorts, generating a complex periodic wave that includes higher harmonics. This waveform distortion is directly correlated with the dynamic evolution of the material's microstructure under large deformations, such as polymer chain disentanglement, micellar breakup, or aggregate network collapse.

Compared to steady-state shear or transient tests, LAOS offers distinct advantages in analyzing complex fluids. While steady-state shear often causes continuous structural breakdown, the periodic loading in LAOS allows the material to undergo alternating deformation-induced destruction and unloading-induced recovery within a single cycle. This dynamic process facilitates the tracking of microstructural network evolution between rupture and reformation. Furthermore, whereas traditional tests typically reflect macroscopic changes in viscosity or modulus, LAOS decouples the nonlinear response into two independent dimensions: nonlinear (e.g., strain hardening/softening, corresponding to network stretching and fracture) and nonlinear viscosity (e.g., shear thinning/thickening, corresponding to interlayer friction). This decoupling aids in identifying specific microstructural mechanisms. By adjusting frequency and amplitude, LAOS can simulate large-deformation dynamic processes such as mastication, pumping, and reciprocating motion within the laboratory, establishing correlations between rheological parameters and actual processing or application performance.

"LAOS Analysis" provides a robust investigative tool for various soft matters and complex fluids. By intimately linking the microstructure and formulation parameters of a system with its macroscopic processing and application performance, LAOS analysis offers critical guidance for revealing structure-mechanics relationships, optimizing formulations, designing processes, and controlling quality. The application domains of this software include, but are not limited to:

(1) Polymer Materials and Processing: Through nonlinear responses, it provides insights into the nonlinear viscoelastic nature of polymer melts/solutions (e.g., quantifying "strain hardening" or "strain softening" behaviors, elucidating the "Payne effect", distinguishing crosslinking types (such as differentiating chemical crosslinks (covalent bonds, exhibiting weak nonlinear responses) from physical crosslinks (hydrogen bonds, etc., exhibiting strong strain-dependent nonlinear responses)), and probing microstructural evolution (e.g., reflecting microstructural rearrangement of polymers in flow fields, providing a basis for optimizing processing techniques like extrusion and injection molding)).

(2) Colloids, Gels, Rubbers, Foams, and Adhesives: Precisely capturing the "yielding" transition of soft matter from solid-like to liquid-like states, as well as the sol-gel transition of colloids and gels; revealing microstructural dynamics, microstructural reorganization under stress, destruction mechanisms, and complex

phenomena like "shear banding" that are inaccessible to SAOS; elucidating the interactions between components (particles) and how they dictate the bulk nonlinear rheological response; simulating the large deformations experienced by adhesives during bonding and peeling, where the derived nonlinear parameters more authentically reflect their adhesive performance; evaluating the effects of different formulations (e.g., reinforcing agents, crosslinkers) on the nonlinear viscoelasticity of rubbers and adhesives; uncovering the relationship between the microstructure and macroscopic nonlinear viscoelasticity and durability of rubber materials; and assessing the long-term stability of emulsions and foams.

(3) Food Processing and Texture: The application of LAOS spans virtually all food categories, providing a powerful research tool for food science. By simulating real-world deformations, it intimately links microstructure, processing, and sensory texture: simulating and differentiating subtle variations induced by processing conditions, predicting actual processing performance, evaluating the effects of processing and storage, and guiding food processing and formulation optimization; quantitatively describing the mechanical behavior of foods during oral processing (e.g., mastication, spreading) and establishing direct correlations with sensory experiences (e.g., simulating the actual eating process, quantifying texture attributes, correlating sensory perception with oral processing, differentiating subtle textural nuances, and sensitively reflecting microstructural changes under stress).

(4) Personal Care Products (Cosmetics and Skincare) R&D: Establishing correlations between rheological parameters and skin feel (e.g., spreadability, thixotropy, cohesiveness, and viscoelasticity); optimizing the stability of various cream/lotion systems such as shampoos, conditioners, toothpastes, lip balms, face creams, hand creams, and sunscreens; quantifying skin feel evaluations and capturing actual user sensations (e.g., simulating the deformation of toothpastes and face creams during "spreading", and establishing the connection between consumer-perceived "texture" and the material's nonlinear rheology).

(5) Biological Fluids: Revealing the anomalous rheological characteristics of non-Newtonian fluids such as blood and mucus, assisting in the study of pathological mechanisms; uncovering complex nonlinear mechanical behaviors under physiological and pathological states, thereby providing critical information for disease diagnosis, drug development, and the understanding of life processes.

(6) Slurries, Coatings, Lubricants, and Inks: Investigating thixotropy and structural recovery, simulating coating and printing processes, predicting film formation and curing behaviors, and evaluating structural collapse during high-shear coating applications and leveling recovery characteristics during subsequent quiescent periods.

(7) Advanced Functional Materials: Evaluating printability and 3D printing performance, quantifying the yield and thixotropic behaviors of 3D printing inks, and resolving the dynamic structural reconstruction of self-healing hydrogels.

1.7 Core Value of the Software

The LAOS Analysis software fills a critical void in the availability of domestic LAOS data analysis tools, providing researchers and engineers with a comprehensive, end-to-end solution that spans from raw data processing to advanced nonlinear rheological analysis. Its core values are manifested in the following aspects:

(1) Highly efficient data processing capabilities and built-in intelligent algorithms automate preprocessing steps—such as data segmentation, noise filtering, phase correction, and Fourier transform—reducing time-consuming manual operations to a matter of minutes;

(2) Integration of diverse analytical methods, including Lissajous curves, Fourier transform rheology, Chebyshev polynomials, stress bifurcation, Sequence of Physical Processes (SPP), and cage modulus. This comprehensive coverage of internationally mainstream LAOS analytical techniques enables users to seamlessly switch between different methodologies;

(3) Intuitive 2D/3D interactive visualization outputs with support for multiple export formats, generating figures that directly meet the publication standards of academic journals;

(4) A combination of a multilingual interface (Chinese, English, and Japanese), customizable plotting styles, and a robust logging feedback mechanism, ensuring a minimal learning curve and an optimal user experience;

(5) User-friendliness for individuals with varying levels of rheological knowledge. Supplemented with typical use cases, even beginners in nonlinear rheology and novice users of LAOS analysis can rapidly master the fundamental methods and skills provided by the software.

1.8 Core Advantages of the Software

Compared to existing analogous software tools or web platforms—such as our prior MATLAB-based algebraic stress bifurcation (ASB) software (https://doi.org/10.60893/figshare.pof.28077395), which solely provides yield stress analysis; The MITlaos software (https://web.mit.edu/nnf/research/phenomena/mit_laos.html) only provides Chebyshev polynomial results; the oreo package (https://github.com/ElsevierSoftwareX/SOFTX-D-21-00058), which is limited to R-based SPP processing; and the Oscillatory Shear Rheology (OSR)-tool (https://edufyx.com/OSR/), which lacks comprehensive mainstream analytical methods and operates as a black-box where users merely upload raw data to passively receive results—as well as overseas data analysis websites inaccessible to domestic users, the LAOS Analysis software exhibits several significant advantages:

(1) The software is packaged as a standalone executable, requiring no Python environment configuration or dependency installation; it is ready to use upon extraction, significantly lowering the barrier to entry;

(2) Superior data compatibility: It fully accommodates rheometers from mainstream instrument manufacturers such as TA Instruments, Thermo Fisher Scientific, Anton Paar, and Netzsch Malvern Panalytical. Notably, it natively supports the proprietary data formats of Thermo HAAKE and TA Instruments, while also being compatible with universal formats like TXT and CSV, effectively resolving the prevalent issue of inconsistent multi-source data formats;

(3) Intelligent preprocessing workflows that automatically identify steady-state cycles, correct phase drift, and eliminate invalid data segments, thereby ensuring the accuracy and reliability of the analytical results;

(4) Modular functional design that allows users to flexibly combine different analytical modules, facilitating effortless multi-file batch processing and comparative analysis;

(5) Continuous optimization: The development team will perpetually refine the software based on user feedback and advancements at the forefront of the field, ensuring its sustained technical sophistication and operational convenience.

1.9 Recommendations for Usage

To fully leverage the functional capabilities of this software, users are advised to

adhere to the following recommendations during practical operation:

(1) Ensure the quality of raw data. It is imperative to verify that the data format is standardized, the encoding is correct (UTF-8 or ANSI), and there are no empty lines or garbled characters, as this is a prerequisite for obtaining accurate analytical results;

(2) Configure parameters rationally. During preprocessing, the experimental frequency must be input accurately; when plotting, parameters such as the Fourier order should be judiciously selected based on specific requirements to prevent overfitting or information loss;

(3) Interpret results prudently. Given that LAOS analysis involves complex rheological theories, it is recommended to critically analyze the results in conjunction with specific material characteristics and experimental conditions, referencing professional literature when necessary;

(4) Regularly back up the automatically saved configuration and plotting style files to prevent the accidental loss of personalized settings;

(5) In the event of operational issues or for suggestions regarding improvement, users should promptly seek technical support via the Help menu or by contacting the development team directly.

1.10 Software Updates

In tandem with the advancement of soft matter science and rheological research, the LAOS Analysis software will undergo continuous iterative upgrades. In the future, the development team plans to incorporate emerging LAOS analytical methods and embed LAOS constitutive models to meet the demands of cutting-edge research. Concurrently, the software will explore the integration of machine learning algorithms to realize intelligent functionalities such as material classification and recognition, anomaly detection, and performance prediction. To enhance collaborative research efficiency, future plans also include the development of a Web-based cloud collaboration version supporting multi-user cooperation and data sharing, alongside the establishment of a repository of typical LAOS data cases to provide standard references [3-5]. Regarding internationalization, additional language versions will be introduced to cater to the needs of a global user base.

By virtue of its robust functionalities, streamlined operation, and comprehensive documentation, the LAOS Analysis software offers an efficient solution for large

amplitude oscillatory shear data analysis. Researchers across disciplines—whether in polymer materials, soft matter, biological fluids, or food processing—can utilize this tool to rapidly acquire profound nonlinear rheological insights and accelerate their research progress. We anticipate that this software will serve as a reliable and indispensable asset for rheologists, collectively propelling the vigorous development of soft matter science.

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Chapter 2: Interface and Basic Functionality

2.1 User Interface Overview

LAOS Analysis features a modular layout, with the interface logically divided into five core functional areas based on operational workflow, encompassing everything from data import and parameter configuration to result output:

(1) Menu Bar: The software provides a comprehensive menu system comprising six main menus. These cover core functionalities including file operations (data folder management and figure export), edit settings (adjustment of plot fonts, styles, and colors), plot generation (rapid creation of various chart types), view control (tab switching and data display), help information (software details and update checks), and system configuration (themes, languages, window settings, and interface personalization). The menu bar supports multilingual switching, keyboard shortcuts, and the persistent saving of custom configurations.

(2) File Information Panel: Displays the name, file path, number of data points, and test identifier for the currently loaded file.

(3) Execution Log Panel: Provides a real-time log of the software's operational status, including analysis progress, algorithm convergence notifications, parameter warnings, and error tracing. In the event of data format mismatches or parameter out-of-bounds errors, the panel highlights corrective suggestions to facilitate rapid troubleshooting.

(4) Visualization Area: Supports multi-window split-screen and overlay comparisons for displaying visual results—such as Lissajous curves and Fourier spectra—alongside data tables. Figures can be exported into various image formats with a single click.

(5) Workspace: Serves as the control center for parameters and view settings, comprising three main modules: plot type selection, data display color configuration, and the parameter settings panel.

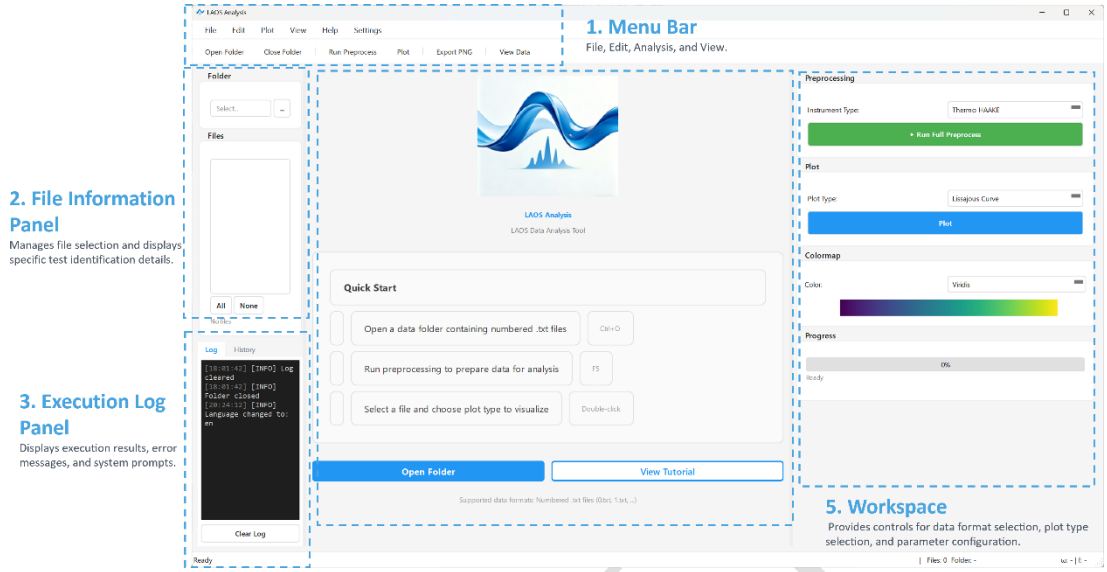


Figure 2. Main interface of the LAOS Analysis software

2.2 Overview of Basic Functions

2.2.1 File Menu

The File menu is primarily designed to manage LAOS experimental data directories and facilitate figure exportation, providing users with streamlined file operations and saving capabilities.

(1) Open Folder: Users can load directories containing LAOS experimental data by clicking the menu item or using the shortcut key (Ctrl+O). Upon selecting the target directory in the dialog box, the system automatically scans for TXT or Excel data files within it and displays the file paths in the left-hand file selector. Concurrently, the successfully opened directory is automatically appended to the "Recent Folders" list for rapid future access.

(2) Close Folder: By utilizing the shortcut Ctrl+W or clicking the corresponding menu item, users can close the currently loaded data directory and clear the file list. It should be noted that this operation simultaneously clears the current plotting area and resets the data preprocessing status; therefore, users are advised to ensure that all critical data has been properly processed or exported prior to execution.

(3) Recent Folders: This feature facilitates rapid access to recently opened data directories. The system automatically tracks and logs historical file paths, enabling seamless transitions between different datasets. These historical records are persistently stored in the local configuration file, remaining accessible even after program restarts.

(4) Export Image File: To accommodate diverse requirements for scientific research and presentation, the File menu offers four figure export formats:

Table 1. Description of Image Export Formats.

Format	Shortcut	Description
Export PNG	Ctrl+S	PNG format; suitable for most applications and supports transparent backgrounds.
Export JPG	None	JPG format; compact file size, suitable for web display.
Export SVG	Ctrl+Shift+S	SVG vector format; ideal for journal publication and high-resolution printing.
Export HTML	None	Interactive HTML 3D plot; supports rotation and scaling within web browsers.

(5) Exit Program: The shortcut Ctrl+Q allows users to safely terminate the application. Upon exiting, the system automatically saves the current preferences (including interface theme, language, and window dimensions) to ensure that the customized workspace is restored upon the next startup.

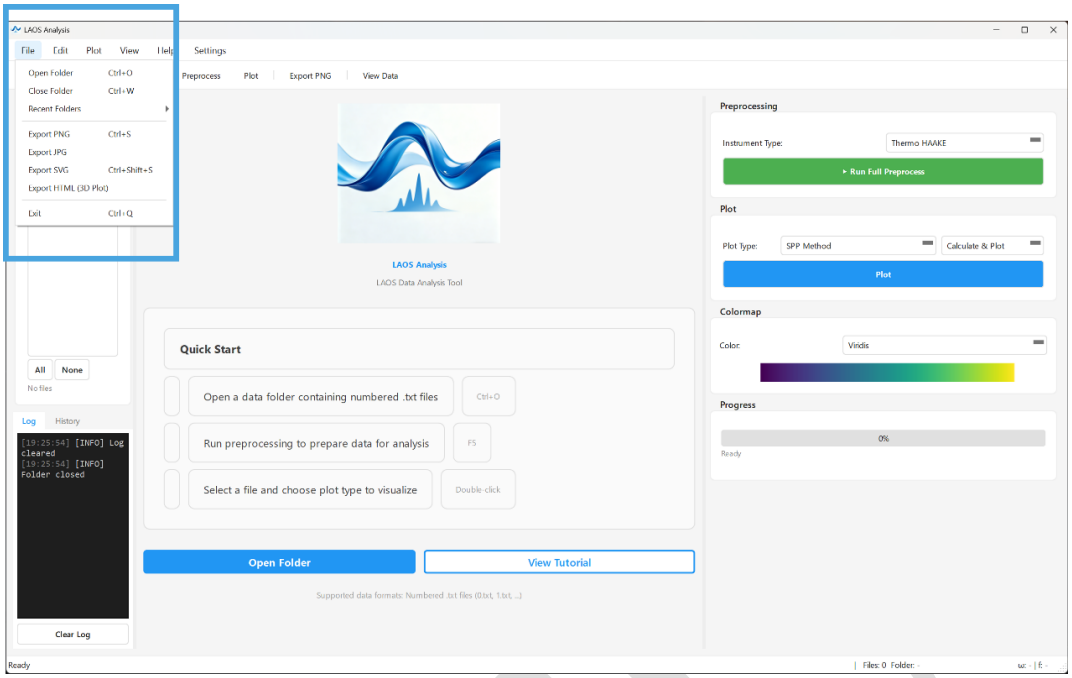


Figure 2. Language switching interface of the software.

2.2.2 Edit Menu

The Edit menu is primarily designed for the customized configuration of visual properties—such as plotting styles, fonts, and colors—enabling users to generate scientific figures tailored to specific requirements.

(1) Plot Font Settings: This function governs the textual attributes within the figures and comprises four sub-menus: Title Font Size (selectable from 12, 14, 16, and 18), Label Font Size (selectable from 10, 12, 14, and 16), Legend Font Size (selectable from 8, 10, 12, and 14), and Global Font Family (selectable from Times New Roman, Arial, and SimHei). All configurations employ a mutually exclusive selection mode, with a checkmark indicating the currently active option. Modified settings take effect immediately and are updated upon the next rendering; furthermore, all configurations are automatically saved to a local file, obviating the need for repeated manual adjustments.

(2) Plot Style Settings: This function allows for the precise control of the visual styles of lines and data markers within the figures, featuring four sub-menus. Specifically, Line Width (selectable from 1 to 10 pixels) directly dictates the stroke thickness in line plots; Marker Size (selectable from 20 to 200) regulates the display dimensions of data points in scatter and overlay plots; Marker Shape (selectable from

Circle, Square, and Triangle) facilitates the intuitive differentiation of distinct data series; and Opacity (selectable from 0.2 to 1.0, where 1.0 denotes full opacity) effectively enhances the discernibility of overlapping data points.

(3) Color Settings: The color settings primarily provide a colormap selection function. The system assigns distinct default colormaps to various plotting functions, which are extensively utilized in scatter and overlay plots. The available colormaps include viridis, jet, jet_r, plasma, cool, hot, rainbow, autumn, summer, and winter. Each colormap possesses distinct characteristics: viridis offers a perceptually uniform color progression, making it the recommended choice for scientific figures; jet is a classic rainbow colormap with vibrant hues, while jet_r denotes its reversed counterpart; plasma features a modern aesthetic suited for high-contrast requirements; and cool and hot render cold and warm tonal palettes, respectively.

(4) Clear and Reset Functions: To facilitate data management and workspace clearance, the Edit menu provides several shortcut operations. Users can execute the "Clear All" function via the shortcut Ctrl+D, which instantly clears the current plotting area, file list, and log panel. If only the execution logs require clearance, the "Clear Log" function can be invoked via Ctrl+L; this operation exclusively empties the bottom log panel without affecting the loaded data or figures. Furthermore, when discarding the current preprocessed data is desired, users can reset the preprocessing data via Ctrl+R. The system will automatically delete the intermediate files generated during preprocessing, thereby enabling users to rapidly revert to the original data state.

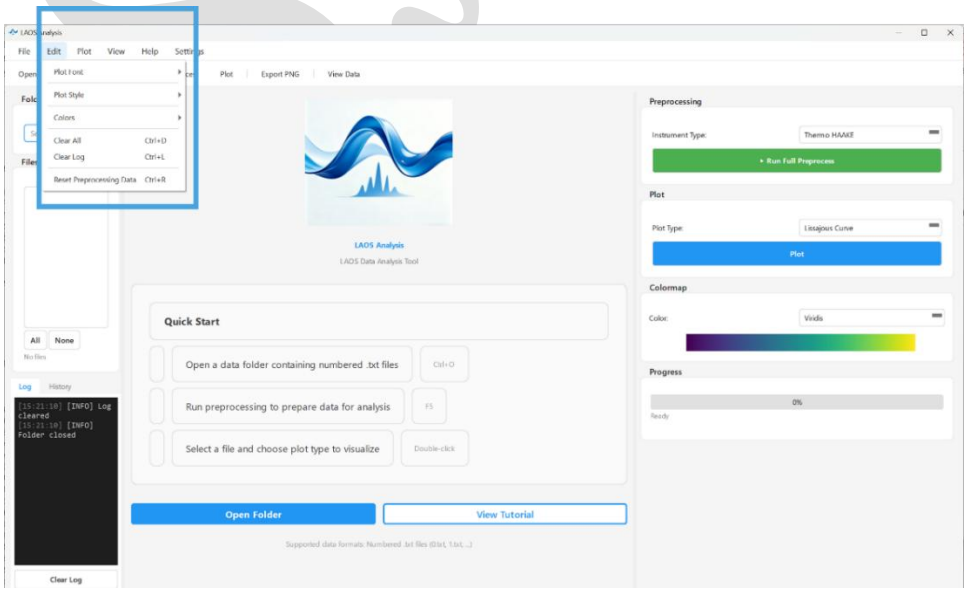


Figure 4. The Edit menu bar of LAOS Analysis.

2.2.3 Plot Menu

The Plot menu provides rapid access to all plotting types and encompasses a variety of plotting functionalities. For a detailed description of the plotting features, please refer to Chapter 4.

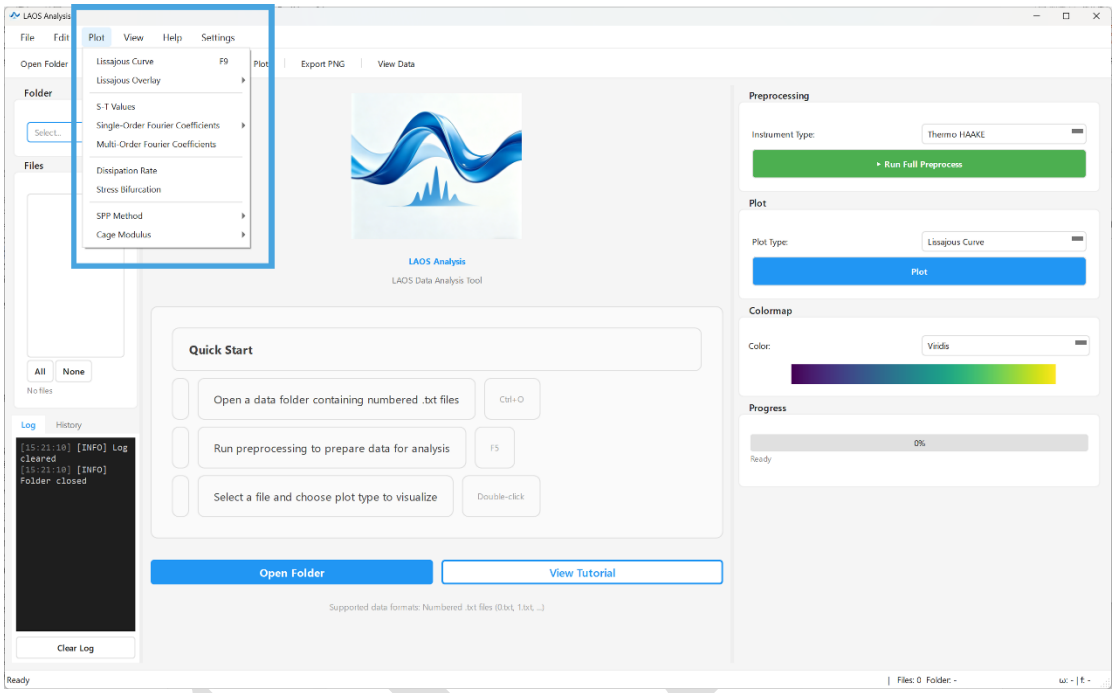


Figure 5. The quick Plot menu bar of LAOS Analysis.

2.2.4 View Menu

The View menu is primarily designed to control the display range of the plotting area, facilitate data inspection, and enable seamless tab switching, thereby providing users with an enhanced interactive visualization experience.

(1) Reset View: Users can rapidly reset the view by pressing the Space key. This function automatically adjusts the axis limits to ensure that the current data is displayed in its entirety within the plotting area. Should users lose their perspective following manual zooming or panning operations, this shortcut enables a one-click restoration to the optimal global observation state.

(2) Display Plot Data: This function invokes a data inspection window that intuitively presents the raw numerical data underlying the current figure. The data

within this window supports one-click copying and exporting to Excel, substantially facilitating underlying data verification and secondary analysis.

(3) Tab Switching: The plotting area comprises two workspaces: the "Analysis Tab" and the "3D Plotting Tab," between which users can seamlessly switch using shortcut keys. Pressing Ctrl+Tab switches to the Analysis Tab for viewing standard 2D/3D analytical figures; conversely, Ctrl+Shift+Tab facilitates rapid transition to the 3D Plotting Tab for in-depth 3D interactive visualization.

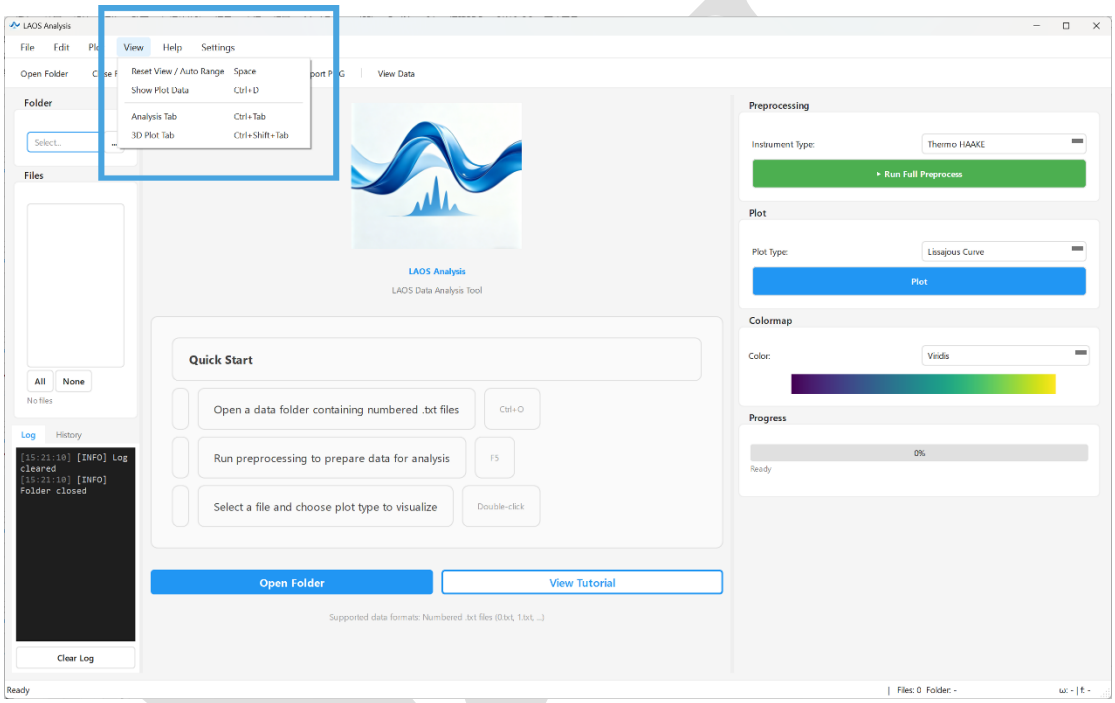


Figure 6. The View menu bar of LAOS Analysis.

2.2.5 Help Menu

The Help menu primarily provides functionalities for querying fundamental software information and checking for version updates, ensuring that users receive timely technical support and software maintenance.

(1) About: This function displays detailed background information regarding the software. Clicking it invokes an information panel encompassing the software name and current version number, an introduction to the development team, a brief overview of the technology stack, and relevant license and copyright statements, thereby

providing users with a comprehensive understanding of the application.

(2) Check for Updates: This function determines whether a newer version of the software is available. Upon execution, the system automatically connects to the update server to compare the local version with the latest server version. If a newer version is detected, the user is prompted to download and upgrade; if the current version is up-to-date, a corresponding notification is displayed, ensuring that the software consistently operates in an optimal state.

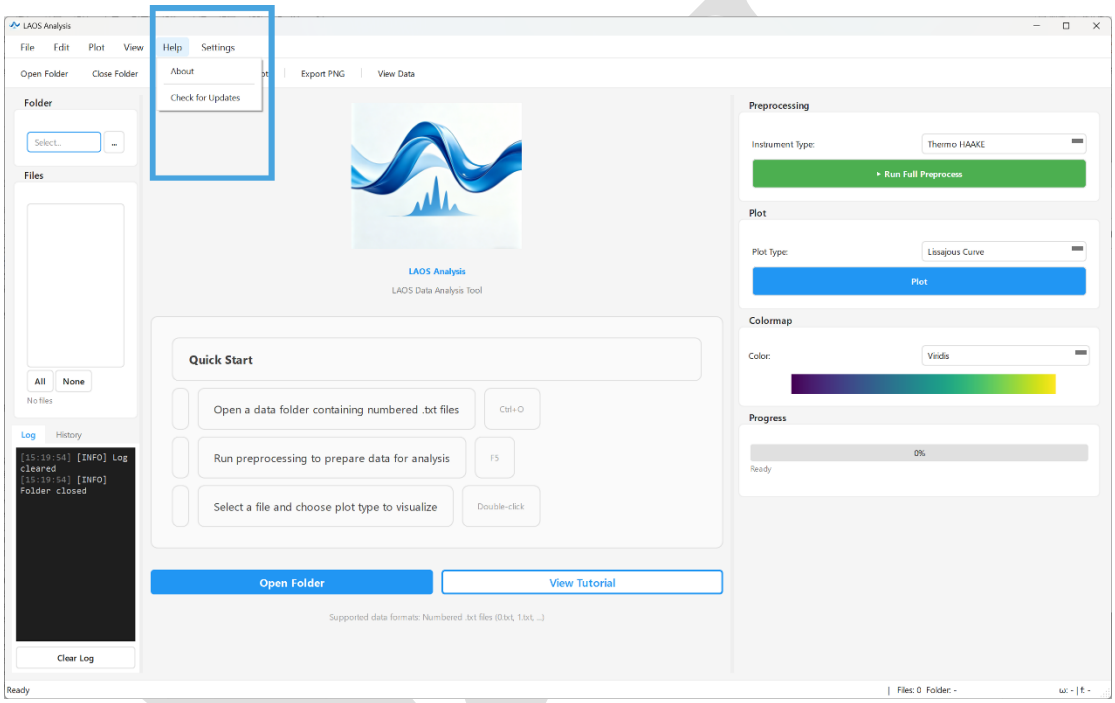


Figure 7. The Help menu bar of LAOS Analysis.

2.2.6 Settings Menu

The Settings menu is primarily utilized for system-level personalized configuration, encompassing multiple dimensions such as themes, language, window dimensions, and interface display, thereby facilitating a tailored and comfortable working environment for users.

(1) Theme Settings: This function controls the overall visual style of the software interface, offering both Light and Dark themes. The Light theme features a bright background, suitable for daytime or well-lit environments; the Dark theme features a

darker background, suitable for nighttime use, effectively reducing eye strain. Upon switching themes, the settings take effect immediately, and all interface elements are updated synchronously.

(2) Language Settings: The system supports a multilingual interface, providing four options: Auto (automatically detects the system language, defaulting to Chinese), English, Chinese, and Japanese. Upon completing a language switch, all menus, buttons, and prompt messages within the software are automatically translated and updated, satisfying the usage requirements of users from diverse linguistic backgrounds.

(3) Window Size Settings: This function adjusts the dimensions of the main window, providing four preset options: Small (1200x800, suitable for small-screen monitors), Medium (1400x900, standard size, recommended), Large (1600x1000, suitable for large-screen monitors), and Maximize (full-screen display, fully utilizing screen real estate).

(4) Interface Font Settings: This function controls the font size of UI elements (such as menus, buttons, and labels). It offers four selectable levels: Small (8pt, compact layout, suitable for small screens), Medium (10pt, standard size, recommended), Large (12pt, large font, suitable for users with visual impairments), and Extra Large (14pt, extra-large font, maximizing readability).

(5) Show/Hide Controls: This function allows users to freely control the display status of interface elements. Users can independently check or uncheck the Toolbar (top toolbar), Status Bar (bottom status bar), and Log Panel (bottom log panel) according to personal preferences. Unchecked elements will be hidden, resulting in a more streamlined and focused workspace.

(6) Export Settings and Reset: To accommodate the varying requirements for image quality and format in scientific figures, the Settings menu provides dedicated export configuration and reset functionalities. The Export Settings primarily configure image export parameters; users can adjust the DPI resolution of exported images to ensure clarity, and set a default export format (such as PNG, JPG, or SVG), thereby eliminating the need for repetitive selection during each export operation. If a restoration to the system's initial state is desired, the "Reset Settings" function can be utilized to clear all personalized configurations with a single click, reverting all parameters to their default values.

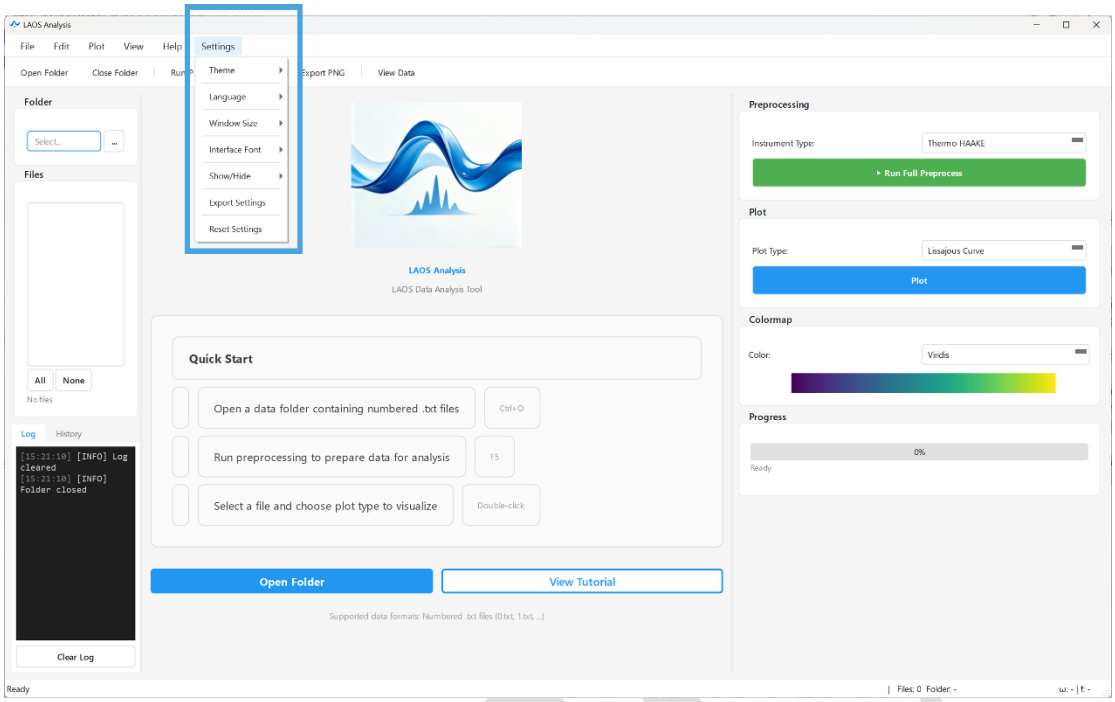


Figure 8. The Settings menu bar of LAOS Analysis.

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Chapter 3 Data Preprocessing

Data preprocessing is fundamental to ensuring the reliability of LAOS analysis results. This module provides automated data cleaning and signal correction tools, designed to rapidly transform raw instrument data into high-quality inputs that conform to the standards of nonlinear rheological analysis.

3.1. Principles of Data Preprocessing

3.1.1 HAAKE MARS III Rheometer

The raw data exported from the HAAKE MARS III rheometer is typically in TXT format, comprising dynamic response records of multiple data points under continuous oscillatory testing. To achieve standardized and modular data analysis, a five-step preprocessing pipeline was designed as follows:

(1) Data decoupling and file splitting: Using "Datapoint no.:" as the identifier, the lengthy raw TXT file is precisely split into multiple independent files, which are sequentially numbered (e.g., 001.txt, 002.txt). For each split single-point file, only the three core characteristic data columns—time, strain, and stress—are retained. This strips away redundant information and facilitates subsequent modular processing.

(2) Denoising and baseline correction: Background noise elimination is performed on the split single-point data. The algorithm extracts the extrema (maximum and minimum values) of the strain and stress data, calculates their mean, and subtracts this value as an offset to achieve efficient baseline correction. This operation aims to eliminate zero-point offsets induced by instrumental system drift and background noise, thereby restoring the true amplitudes of the dynamic responses.

(3) Phase alignment and correction: To eliminate initial phase differences among data points, the strain peak positions are uniformly located and aligned. The algorithm accurately identifies the global peak of the strain data and shifts it to the quarter-length position of the data array (i.e., the theoretical peak position of a standard sine wave) via circular shifting. This operation ensures the consistency of the initial phase across all data, establishing a benchmark for subsequent superposition comparisons and nonlinear feature extraction.

(4) Analytical calculation of shear rate: By incorporating the user-defined oscillation frequency, a sinusoidal function is fitted to the strain trajectory to

analytically determine the strain rate (shear rate). The fitting model is defined as: $\gamma(t) = A \cdot \sin(2\pi f \cdot t + \phi) + C$ (where f is the frequency in Hz). The high signal-to-noise ratio (SNR) shear rate is then obtained via differentiation: $\dot{\gamma}(t) = 2\pi f \cdot A \cdot \cos(2\pi f \cdot t + \phi)$. To enhance algorithmic robustness, the system automatically degrades to a numerical gradient method for approximate estimation if data distortion causes the fitting to fail.

(5) Frequency-domain transformation and harmonic extraction: A Fast Fourier Transform (FFT) is applied to the stress data following time-domain preprocessing, converting it into the frequency domain to deeply mine nonlinear characteristics. The algorithm focuses on extracting the coefficients of the first 30 higher-order harmonics. The output encompasses the amplitude (I_n), real part, and imaginary part of each harmonic, which are archived by index into a fourier_coefficients subfolder for subsequent nonlinear rheological analyses (e.g., Chebyshev coefficient decomposition).

3.1.2 Data Processing Pipeline for TA Instruments

Compared to the TXT format used by HAAKE, the Excel files exported by TA Instruments rheometers adopt a multi-worksheet architecture, where each worksheet independently records multi-cycle oscillatory shear experimental data at a specific strain amplitude. The preprocessing pipeline for this format focuses on the precise extraction of periodic features and the reconstruction of steady-state waveforms, with the specific steps as follows:

(1) Intelligent worksheet scanning: The system automatically traverses the Excel file to accurately identify all worksheets prefixed with "Sine Strain -" and extracts the corresponding strain indices to establish a data index.

(2) Dual detection of oscillation period: An FFT spectral analysis is performed on the strain sequence within the worksheet to identify the dominant frequency and derive the period length. To ensure absolute reliability of period identification, a peak detection algorithm is introduced, utilizing the median of peak intervals as a cross-validation benchmark to prevent period misidentification caused by data distortion.

(3) Precise determination of period boundaries: Instead of employing crude truncation methods, the algorithm precisely anchors the start and end boundaries of each complete oscillation cycle based on the ascending/descending intersection points between the strain data and its mean line, thereby ensuring the completeness of period

extraction.

(4) Period segmentation and steady-state averaging: The long-sequence continuous data is segmented into independent periods according to the defined boundaries. The algorithm features an adaptive steady-state identification mechanism, preferentially screening stable periods without amplitude drift (relative error < 5%). Finally, an ensemble average is performed on the waveforms of the target periods to filter out random fluctuations and extract a highly representative single-period standard waveform.

(5) Standardized post-processing: The extracted single-period data is seamlessly fed into the unified data processing pipeline, sequentially executing steps identical to the HAAKE workflow: "Denoising and baseline correction", "Phase alignment and correction", "Analytical calculation of shear rate", and "Frequency-domain transformation and harmonic extraction". This ensures high comparability of analytical results across data derived from different instruments.

3.2. Operational Workflow and Verification Specifications

Currently, the system is fundamentally compatible with the LAOS exported data formats from two mainstream rheometers: Thermo HAAKE and TA Instruments. It automatically identifies and extracts the three core data columns—time, strain, and stress—and supports the automated calculation and supplementation of strain rate.

To ensure analytical stability, users are advised to verify prior to import that the source data is free of blank rows and unconventional characters (e.g., encoding errors caused by mojibake). The module is specifically designed for strain-controlled LAOS tests. The built-in algorithms can automatically perform phase drift correction induced by instrument inertia and feature breakpoint detection capabilities to accurately eliminate invalid data segments that have not reached steady-state oscillation.

The standard operational workflow is as follows:

(1) **Data file selection:** Specify the rheometer brand/data type within the interface, and select the raw data file to be imported for processing.

(2) **Data preprocessing:** Click the "Preprocess" button, and the system will automatically execute core cleaning steps in sequence, including steady-state period segmentation, high-frequency noise filtering, and phase correction.

(3) **Result verification:** Upon completion, review the file information area and the

execution log. If successful, the system will automatically split and generate corresponding standard TXT data files categorized by different strain amplitudes. If the process fails, the log area will display error tracking messages in red (e.g., no steady-state period detected, format mismatch) to facilitate rapid troubleshooting.

3.3. Preprocessing Output Data

Upon successful preprocessing, the system outputs standardized data files categorized by strain amplitude (γ_0), primarily comprising the following two core outputs:

(1) Raw data corresponding to different strain amplitudes: This file contains four columns of data, which from left to right are: time/sampling point index, strain, stress, and strain rate. It retains the true time-domain response waveforms after denoising and steady-state truncation, serving as the data foundation for plotting Lissajous curves.

(2) Frequency-domain Fourier coefficients: This file contains three columns of data, which from left to right are: relative harmonic intensity (I_n), higher-order loss modulus (G_n''), and higher-order storage modulus (G_n'). The system defaults to calculating and retaining up to the 30th-order Fourier coefficients, comprehensively covering the high-order harmonic information required for nonlinear analysis (e.g., characteristic odd harmonics such as I_3/I_1 and I_5/I_1) to quantify the nonlinear viscoelastic responses of materials.

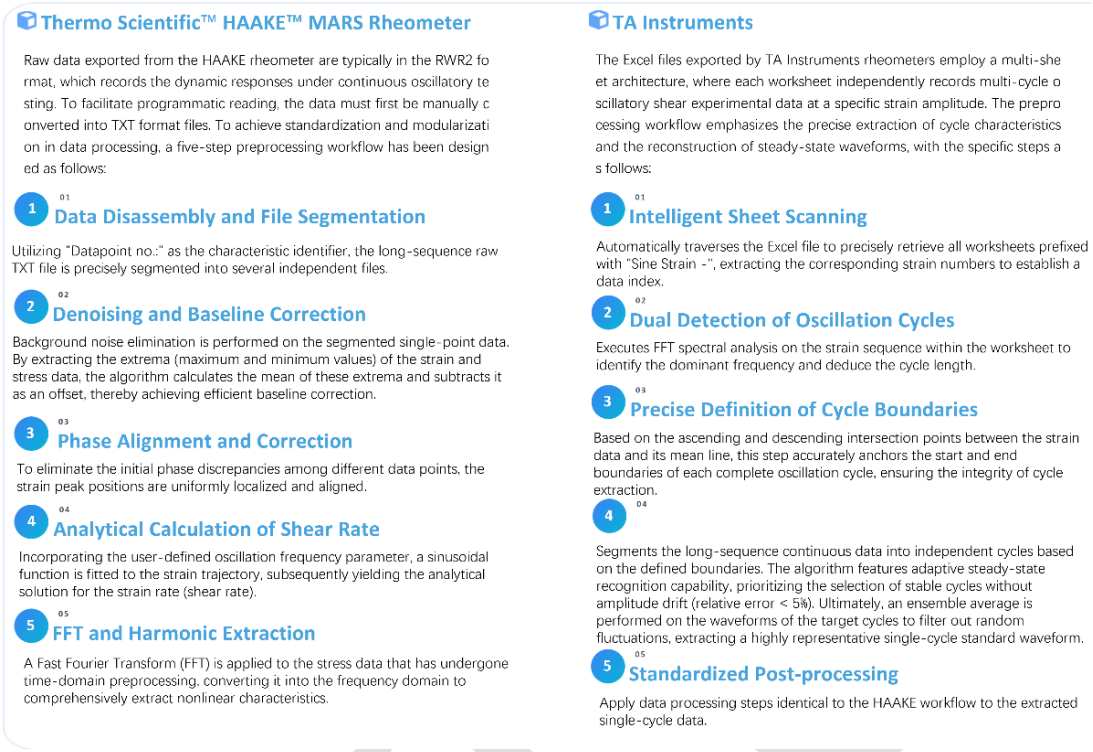


Figure 9. LAOS Analysis data preprocessing workflow.

Chapter 4 Plotting Functions

4.1 Lissajous Curves

4.1.1 Basic Principle

The Lissajous curve method serves as an intuitive and efficient graphical qualitative evaluation tool for characterizing the viscoelastic behavior of materials in rheology. By plotting the closed-loop curves of oscillatory stress versus strain (or strain rate), this method transforms complex dynamic rheological data into intuitive geometric trajectories. Consequently, it enables researchers to rapidly assess the structural evolution, yielding characteristics, and energy dissipation mechanisms of materials [1].

This method primarily encompasses two core types of plots: the elastic Lissajous plot, which emphasizes the dynamic response between stress and strain to reveal the elastic characteristics and nonlinear distortions of the material; and the viscous Lissajous plot, which focuses on the coupling relationship between stress and strain rate to delineate the viscous flow behavior. The rheological characteristics of the material can be rapidly identified by observing the geometric morphology of the curves. Within the linear viscoelastic (LVE) region, the stress and strain exhibit a proportional relationship, resulting in a regularly shaped, tilted ellipse. As the material transitions into the nonlinear region, the stress response generates higher-order harmonics, causing the curve to undergo twisting and distortion. For instance, a change in the slope of the ellipse's major axis reflects a variation in the modulus, whereas the emergence of a plateau often indicates the onset of yielding or plastic flow within the material. Furthermore, the area enclosed by the loop possesses a distinct physical significance, directly corresponding to the energy dissipated by viscous friction within a single oscillation cycle.

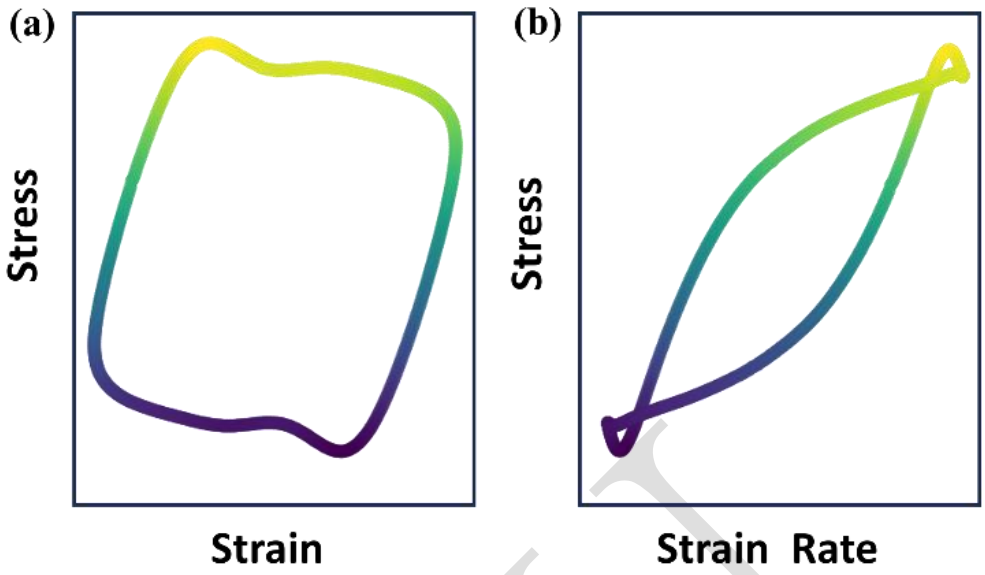


Figure 10. (a) Elastic and (b) viscous Lissajous plots for 1% xanthan gum (XG) at a strain amplitude of 150%.

4.1.2 Operational Procedure

The specific procedural steps for plotting Lissajous curves are as follows:

(1) Step 1: Module invocation. Navigate to the main interface of LAOS Analysis. Within the Plot module, select the "Lissajous Curve" function under the plot type options.

(2) Step 2: Data loading. Input the index number corresponding to the target *.txt* file into the file index field. The system will automatically parse the data and dynamically render the Lissajous curves in real time.

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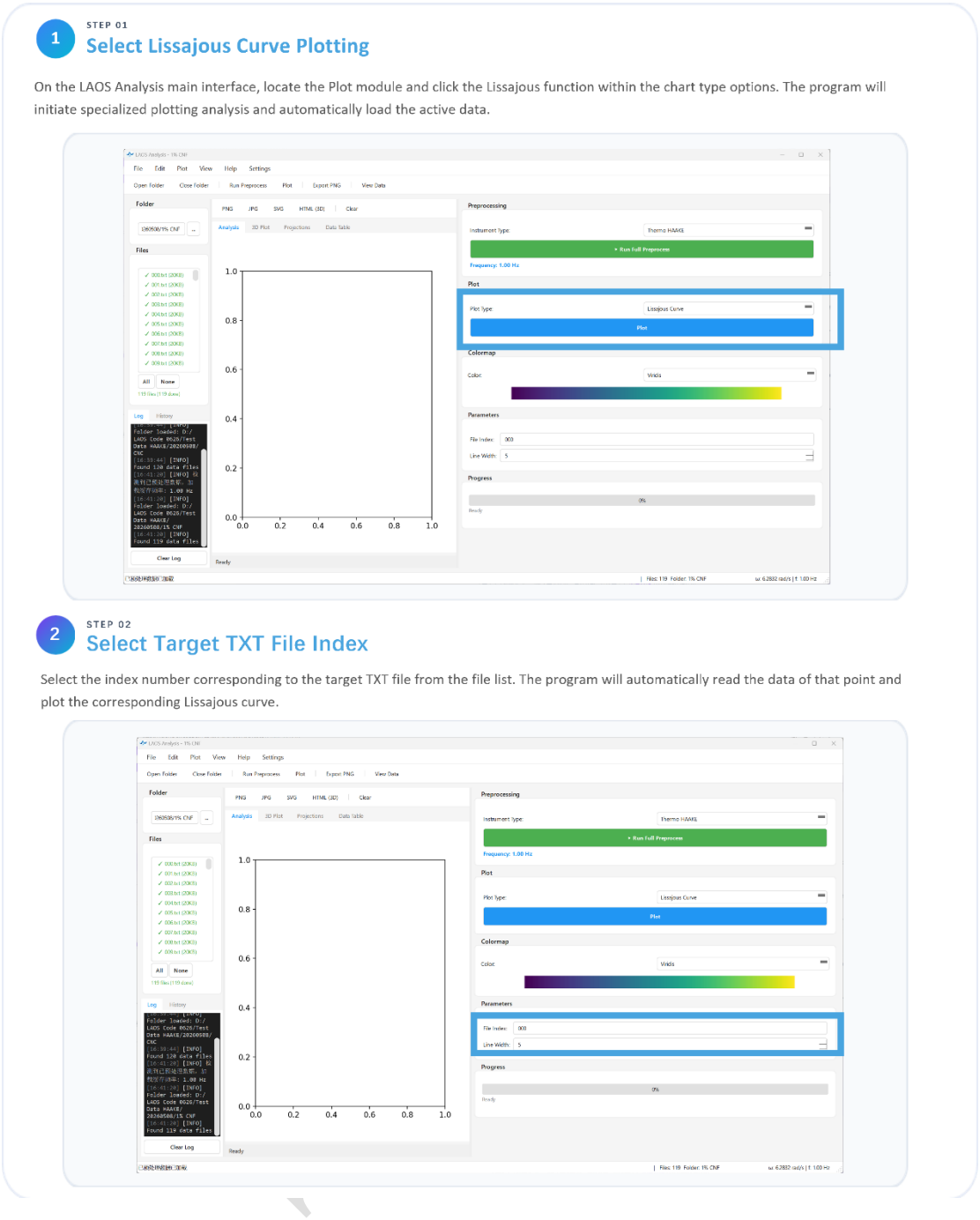


Figure 11. Configuration steps for the Lissajous curve plotting program.

4.1.3 Output Results

Based on the selected color settings and target file, the system generates the following outputs:

(1) 3D View Window: Displays a three-dimensional rendering of the Lissajous

curves. It supports interactive adjustment of the viewing perspective via mouse dragging and allows for one-click export into PNG, SVG, or interactive HTML formats.

(2) Projection Window: Separately displays the elastic and viscous Lissajous plots, facilitating the independent analysis of the elastic and viscous responses.

(3) Data Table Window: Presents the raw data of stress, strain, and strain rate for the selected file in real time. It supports one-click export to Excel format or direct copying to the clipboard for subsequent data processing.

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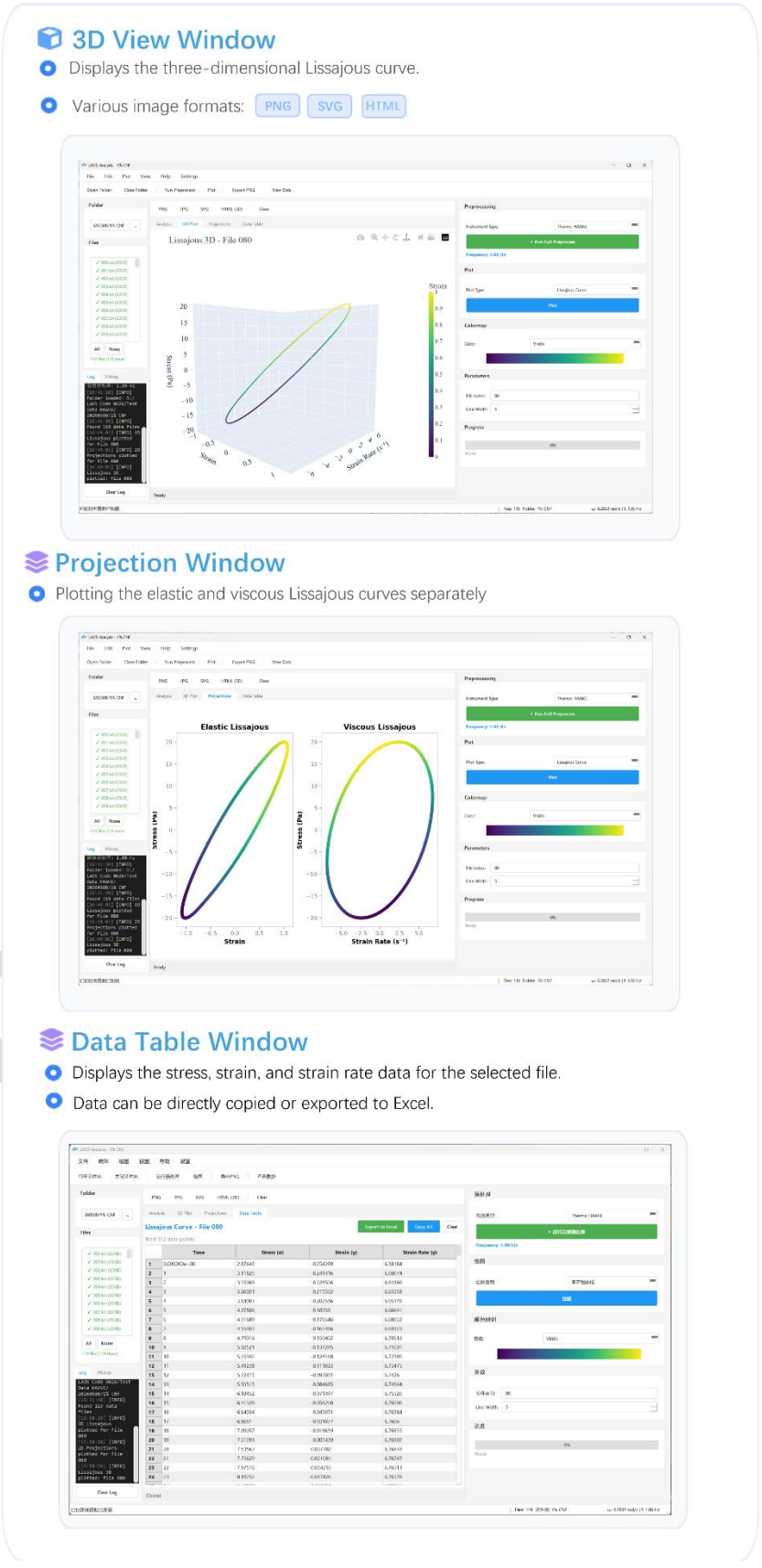


Figure 12. Output of the Lissajous curve plotting program.

4.2 Lissajous superposition curves

4.2.1 Fundamental principles

The superposition of Lissajous loops is an advanced graphical qualitative and semi-quantitative evaluation method derived from the traditional Lissajous curve approach. By plotting a series of Lissajous loops obtained at various strain amplitudes within a single coordinate system, a superimposed plot is generated. This provides a highly intuitive visualization of the entire dynamic response process of the material as it transitions from the linear to the nonlinear region.

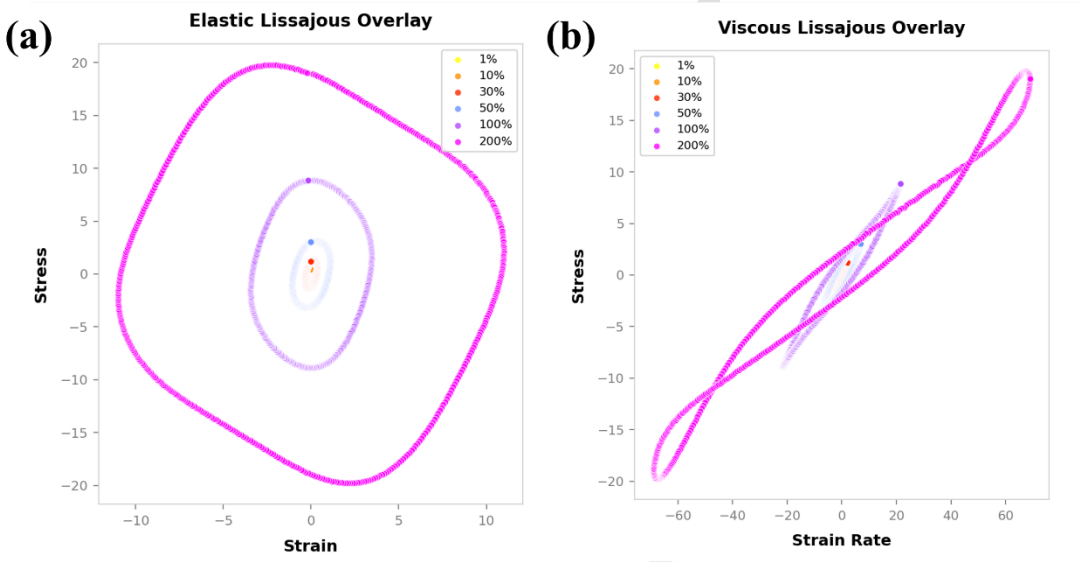


Figure 13. Lissajous superposition curves of a 1% cellulose nanocrystal (CNC) suspension at various strain amplitudes.

4.2.2 Operational procedure

The operational procedure for generating Lissajous superposition curves is as follows:

(1) Step 1: Module selection. Select the "Lissajous superposition" module from the plot types. This module enables the generation of 3D superimposed Lissajous loops, as well as separate superimposed elastic and viscous Lissajous loops.

(2) Step 2: Data loading. Input the identifiers of multiple files, separated by commas (,).

(3) Step 3: Label configuration. Assign individual labels to each file, with options

to save and load label configurations.

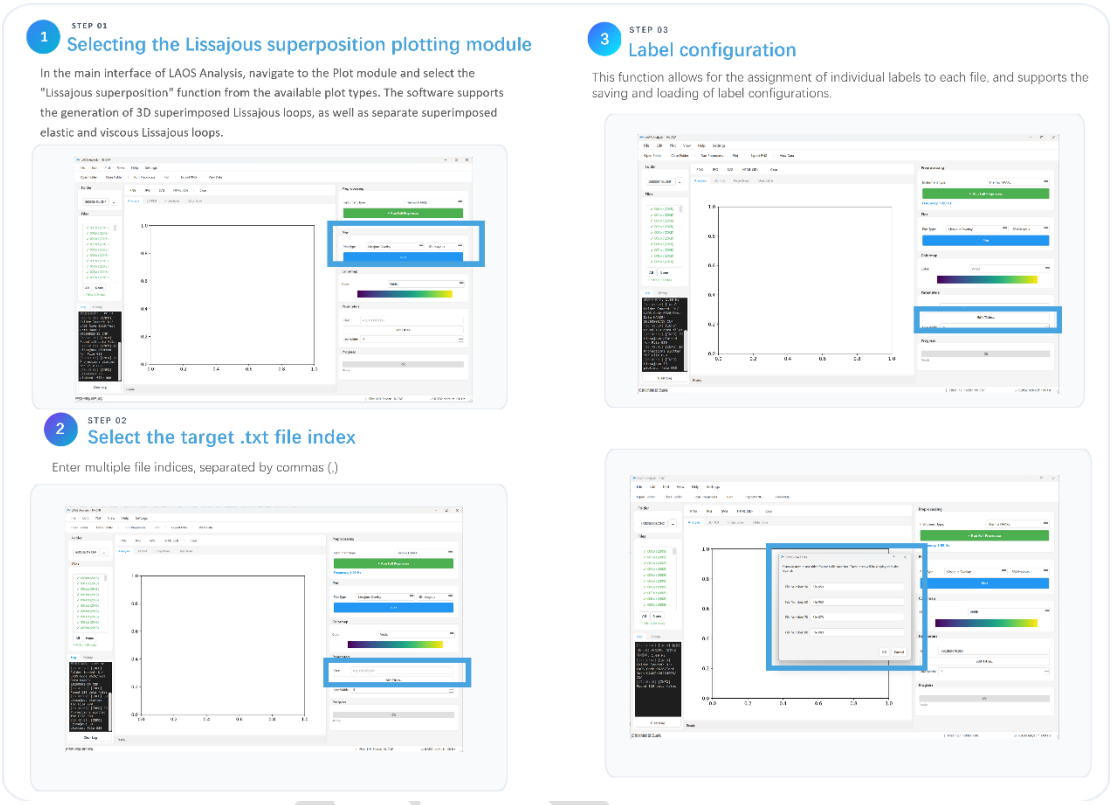


Figure 14. Configuration steps for the Lissajous superposition curve program.

4.2.3 Output results

Based on the selected color schemes and target files, the system generates the following outputs:

(1) 3D view window: Renders the 3D superimposed Lissajous curves, allowing interactive rotation of the viewing angle via mouse dragging. The plot can be directly exported in PNG, SVG, or interactive HTML formats.

(2) Projection window: Displays the superimposed elastic and viscous Lissajous curves separately.

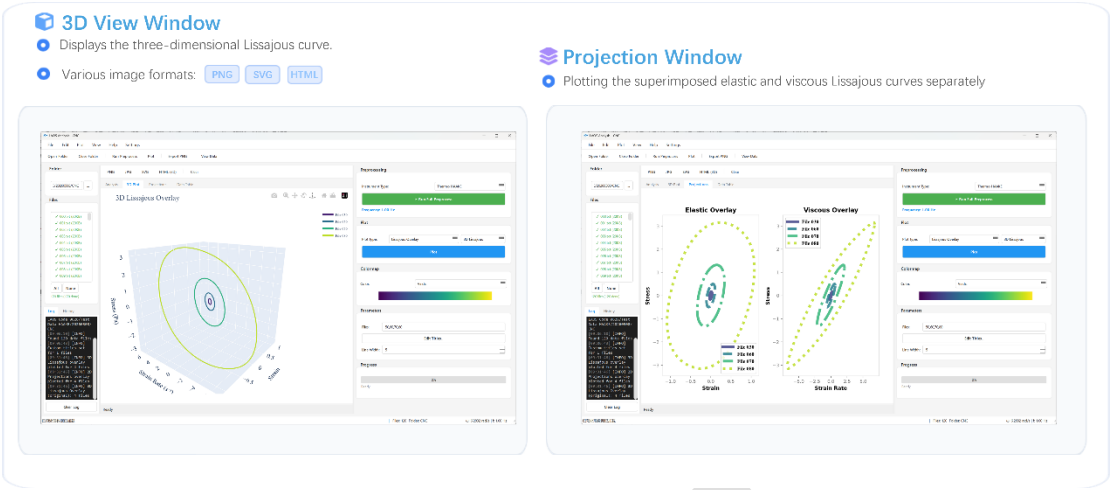


Figure 15.Output of the Lissajous superposition curve program.

4.3 Dissipation ratio

4.3.1 Fundamental principles

By calculating the enclosed area of the Lissajous loops at various strain amplitudes and converting it into the dissipation ratio, an evolution curve illustrating the dissipation ratio as a function of loading conditions can be plotted. This allows for the precise quantification of energy loss and structural evolution characteristics of the material during dynamic deformation.

The dissipated energy per unit volume per cycle, E_d, can be expressed as the enclosed area of the stress-strain curve:

$$E_d = \int_0^{2\pi/\omega} \sigma(t) d\gamma(t) = \pi \gamma_0^2 G_1'' \tag{1}$$

The dissipation ratio, ϕ, is obtained by normalizing the actual dissipated energy by the perfectly plastic dissipation:

$$\phi \equiv \frac{E_d}{(E_d)_{pp}} = \frac{\pi \gamma_0 G_1''}{4\sigma_{max}} = \begin{cases} \to 1 \; Perfet\ Plastic \\ \to \frac{\pi}{4} \approx 0.785 \; Newtonian \\ \to 0 \; Purely\ Elastic \end{cases} \tag{2}$$

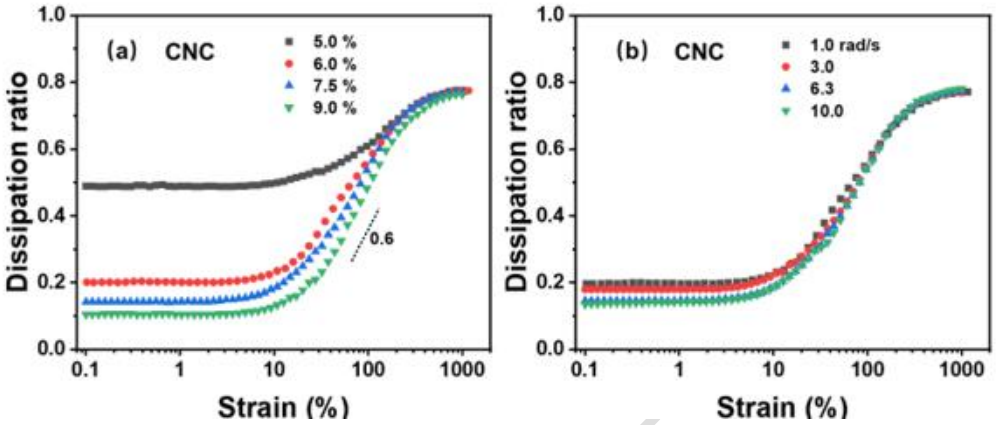


Figure 16. Variation of dissipation ratio of CNC at different concentrations (a) and frequencies (b) [6].

4.3.2 Operating Procedure

By selecting the dissipation ratio function module within the plot type options, the program will automatically load the current data for plotting and analysis.

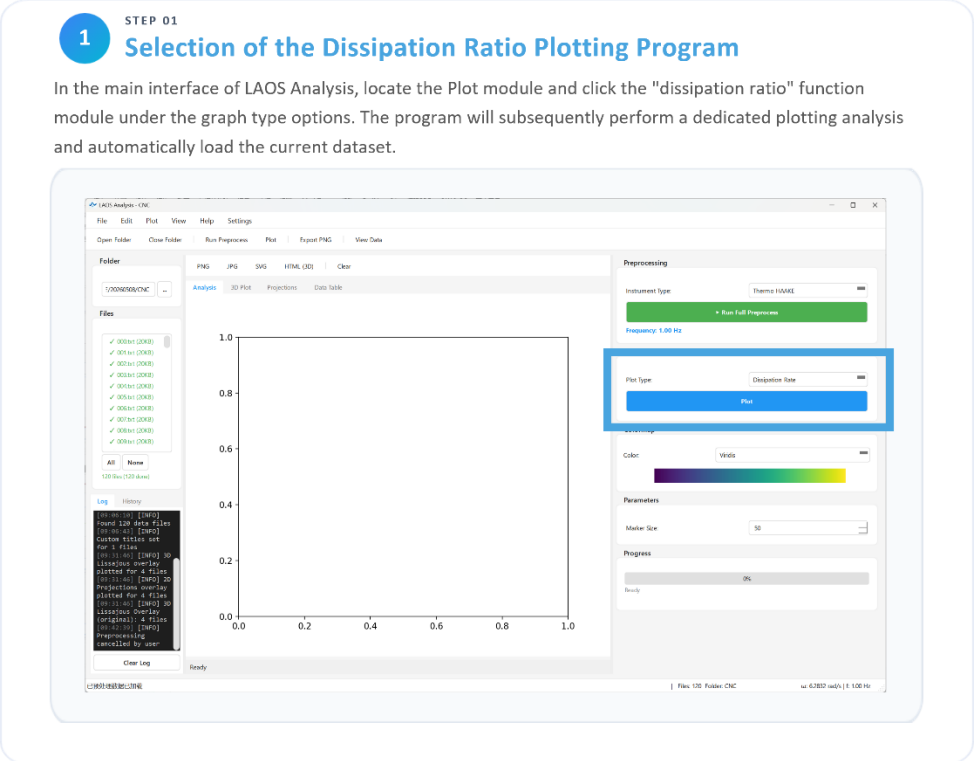


Figure 17. Program settings for the dissipation ratio curve plotting module.

4.3.3 Output Results

Based on the selected colors and target files, the system will generate the following output results:

(1) Analysis window: Displays the curve of the dissipation ratio as a function of strain amplitude.

(2) Data Table window: Provides two columns of data, corresponding to strain and dissipation ratio, respectively.

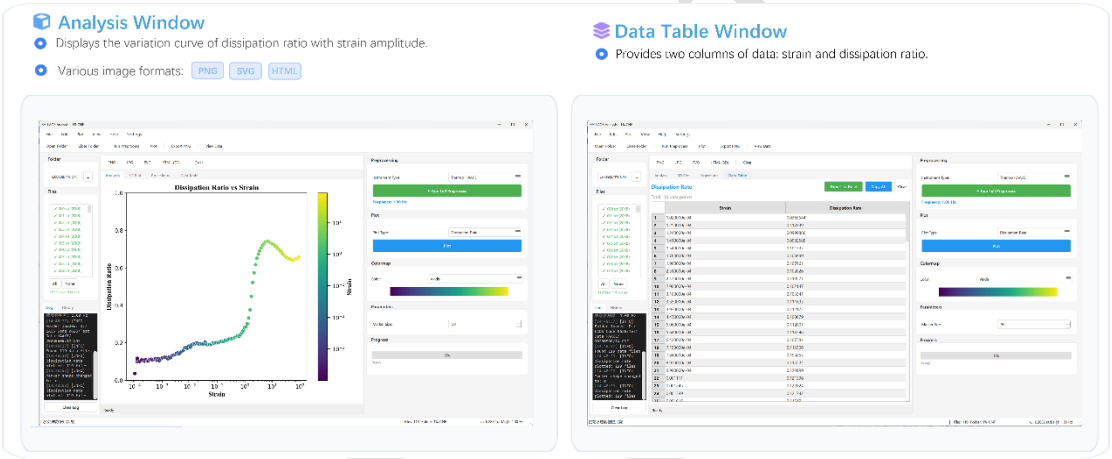


Figure 18. Result display of the dissipation ratio program.

4.4 Fourier Coefficients

4.4.1 Basic Principles

Fourier-transform rheology is a quantitative analysis method formed by introducing the mathematical tool of Fourier transform into the field of rheology, specifically designed to resolve the nonlinear viscoelastic responses under large-amplitude oscillatory shear (LAOS) [1, 7].

In the small-amplitude oscillatory shear (SAOS) regime, the stress response of the material remains at the same frequency as the input strain, presenting a perfect sinusoidal wave. However, when the deformation enters the large-amplitude regime (LAOS), the stress waveform distorts and is no longer a single simple harmonic oscillation. The core logic of Fourier-transform rheology is to transform this complex and distorted stress signal from the time domain into the frequency domain for

decomposition:

$$\sigma(t) = \sum_{n=1}^{\infty} I_n \sin(n\omega t + \phi_n) = \gamma_0 \sum_{n=1}^{\infty} (G_n'' \cos n\omega t + G_n' \sin n\omega t) \quad (3)$$

Specifically, based on the principle of Fourier series, the nonlinear stress response can be expanded as a linear superposition of the fundamental frequency (the first harmonic, corresponding to the driving frequency ω) and its higher harmonics (primarily odd multiples such as the third harmonic 3ω and the fifth harmonic 5ω). Within this frequency-domain framework, the fundamental frequency component represents the foundational linear viscoelastic response of the material, whereas the emergence and intensity of the higher harmonics serve as a direct 表征 (characterization) of the material's nonlinear mechanical behavior.

Among various frequency-domain parameters, the relative intensity ratio of the third harmonic to the first harmonic (I_3/I_1) is the most fundamental and commonly used quantitative metric in Fourier-transform rheology for measuring the severity of nonlinear distortion. To further decouple the influence of strain amplitude and describe the nonlinear characteristics of materials more essentially, researchers introduced the nonlinear parameter Q. The Q value is defined as the ratio of the normalized third-harmonic intensity to the square of the strain amplitude, i.e., $Q = I_{3/1}/\gamma_0^2 = (I_3/I_1)/\gamma_0^2$. In the medium-amplitude oscillatory shear (MAOS) regime, I_3/I_1 is typically proportional to γ_0^2, at which point the Q value approaches a constant (denoted as Q_0). This constant is determined solely by the intrinsic properties of the material and is independent of the applied strain magnitude; thus, it is regarded as the "fingerprint" of the material's intrinsic nonlinear response. By analyzing the Q value, the degree of nonlinearity under different strain conditions can be compared more accurately, significantly enhancing the discriminative sensitivity to microstructural evolution (such as polymer branching, disentanglement, network breakdown, etc.) [8, 9].

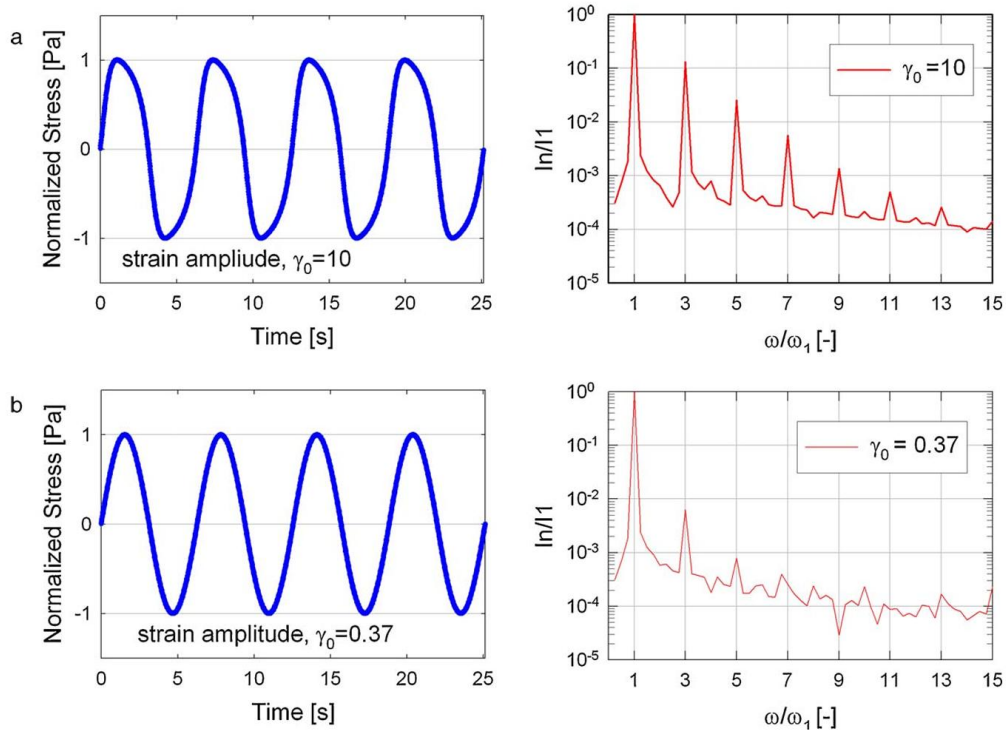


Figure 19. Normalized stress curve and Fourier transform spectrum of a 1 wt% hyaluronic acid aqueous solution at a frequency of 1 rad/s.

4.4.2 Operating Procedure

4.4.2.1 Single-Order Fourier Coefficient

The specific operational steps for plotting the single-order Fourier coefficient curve are as follows:

(1) Step 1 Module Activation: Enter the main interface of LAOS Analysis, and within the Plot module, click the "Single-Order Fourier Coefficient" function module under the graph type options.

(2) Step 2 Parameter Setting: Set the order of the Fourier coefficient in the parameter module; the default order is 3.

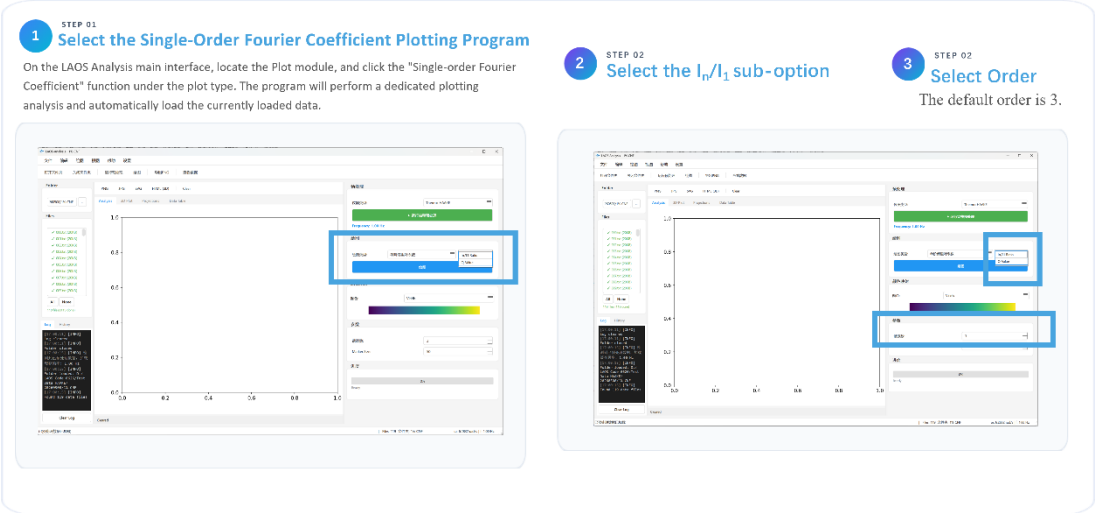


Figure 20. Step settings for the single-order Fourier coefficient plotting program.

4.4.2.2 Q-Value Plotting

The specific operation steps for plotting the curve are as follows:

(1) Step 1 Module invocation: Go to the LAOS Analysis main interface, and in the Plot module, click the "Single-order Fourier Coefficient" function under the plot type option.

(2) Step 2 Select the Q-value sub-option.

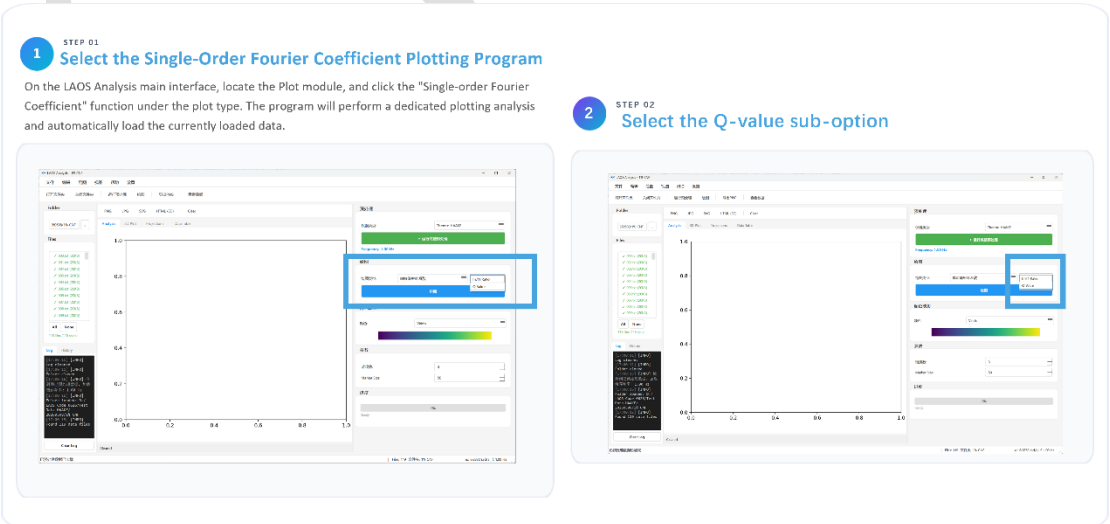


Figure 21. Step settings for the Q-value plotting program.

4.4.2.3 Multi-Order Fourier Coefficients

The specific operation steps for plotting the multi-order Fourier coefficient curves are as follows:

(1) Step 1 Module invocation: Go to the LAOS Analysis main interface, and in the Plot module, click the "Multi-order Fourier Coefficients" function under the plot type option.

(2) Step 2 Data loading: In the multi-file number input box, type the corresponding numbers of the files to be overlaid and displayed. Note that the numbers must be strictly separated by English commas.

(3) Step 3 Parameter setting: In the parameter configuration area, set the maximum number of harmonics (i.e., the order of the Fourier expansion) according to the actual fitting and analysis requirements. The system allows a maximum setting of 30 orders.

(4) Step 4 Label setting: Enter a custom curve title for each file number, which will be displayed in the legend.

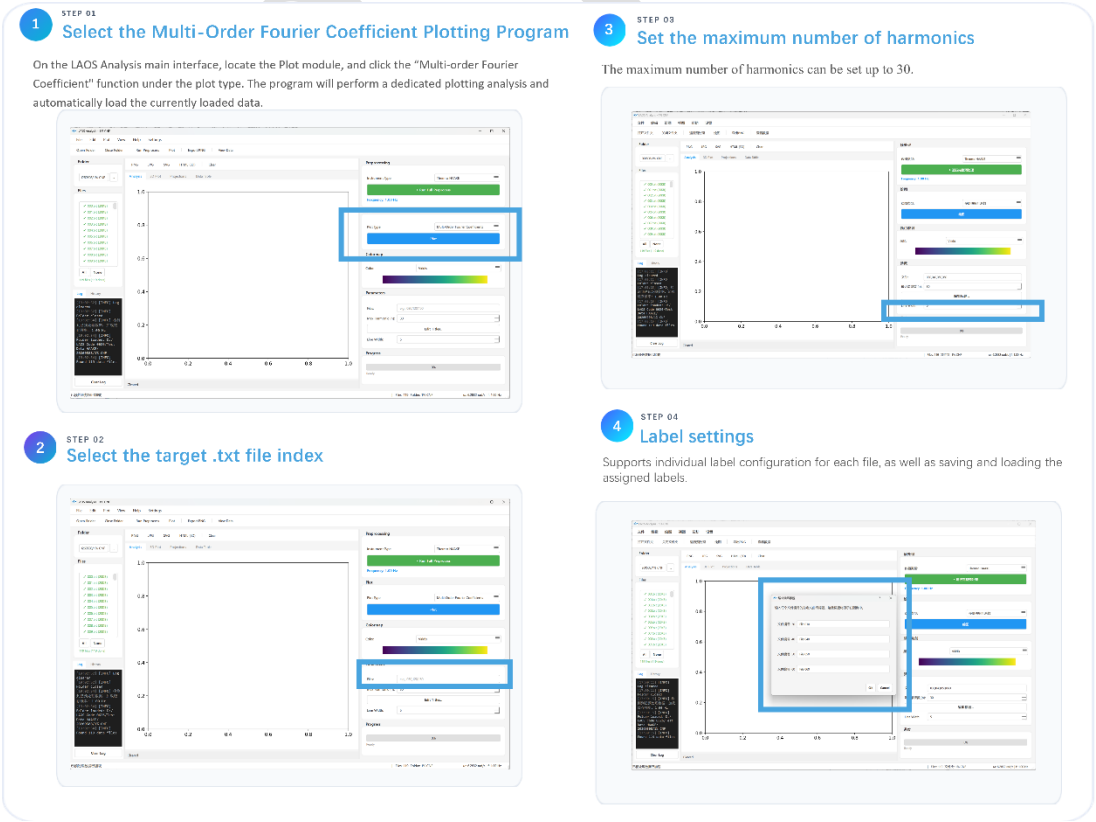


Figure 22. Step settings for the multi-order Fourier coefficient plotting program.

4.4.3 Output Results

4.4.3.1 Single-Order Fourier Coefficient

Displays the curve of the single-order Fourier coefficient changing with strain.

(1) Analysis window: Displays the curve of I_n/I_1 versus strain amplitude.

(2) Data Table window: Provides two columns of data: strain and I_n/I_1.

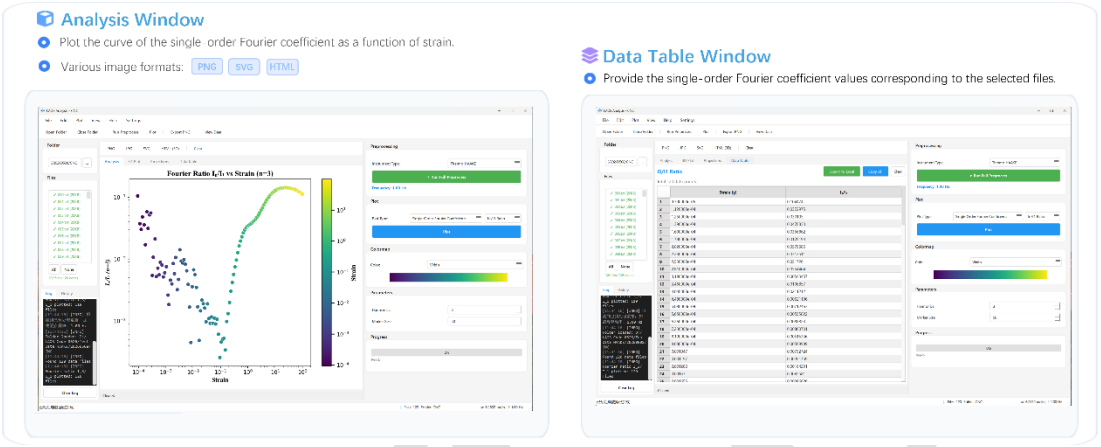


Figure 23. Results display of the single-order Fourier coefficient plotting program.

4.4.3.2 Q-Value

Displays the curve of the Q-value changing with strain.

(1) Analysis window: Displays the curve of the Q-value versus strain amplitude.

(2) Data Table window: Provides two columns of data: strain and Q-value.

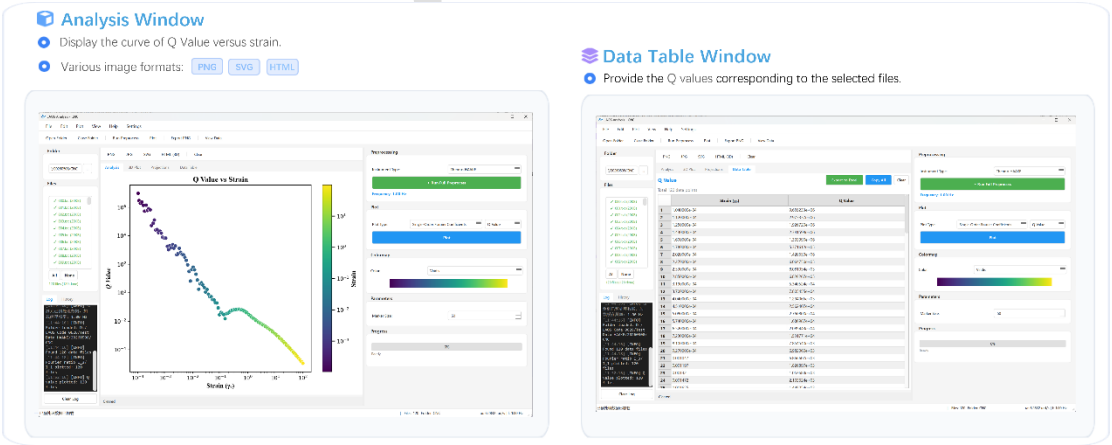


Figure 24. Results display of the Q-value plotting program.

4.4.3.3 Multi-Order Fourier Coefficients

Displays 3D curves of Fourier coefficients of different orders changing with strain, supporting free rotation and scaling.

(1) 3D View window: Presents a stereoscopic 3D view of the multi-order Fourier coefficients, supports interactive changing of the observation perspective via mouse dragging, and allows one-click export to PNG, SVG, or HTML interactive formats.

(2) Data Table window: Displays the multi-order Fourier coefficient values corresponding to each file.

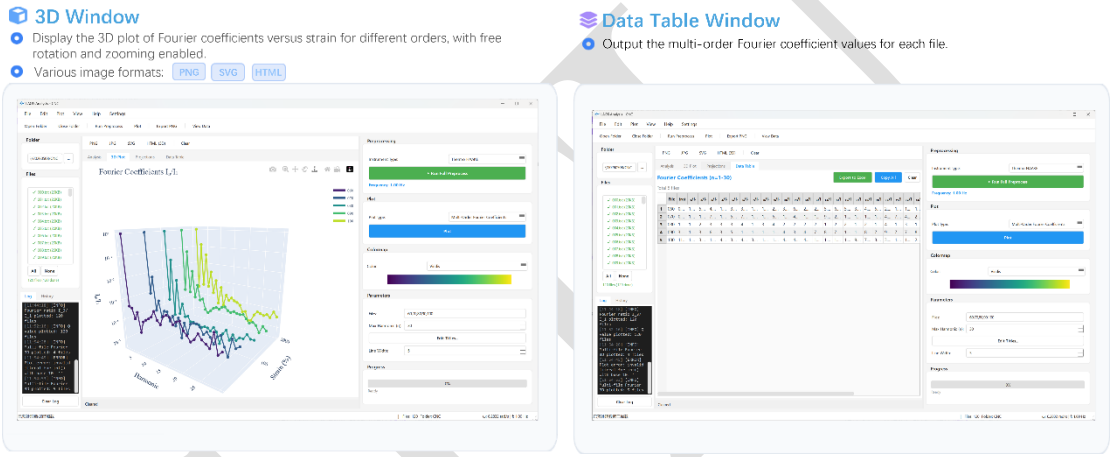


Figure 25. Results display of the multi-order Fourier coefficient plotting program.

4.5 ST Values (Chebyshev Polynomials)

4.5.1 Basic Principles

The Chebyshev polynomial method is another core mathematical framework for analyzing the nonlinear response in LAOS. Unlike Fourier transform rheology, which decomposes the signal in the frequency domain, this method is based on strain and strain rate, expanding the nonlinear stress response through orthogonal polynomials [10].

$$\sigma'(x) = \gamma_0 \sum_{n:odd} e_n(\omega, \gamma_0) T_n(x) \quad (4)$$

$$\sigma''(y) = \dot{\gamma}_0 \sum_{n:odd} v_n(\omega, \gamma_0) T_n(y) \quad (5)$$

Its fundamental logic is to decompose the total stress into elastic and viscous components, fitting them separately using Chebyshev polynomials of strain and strain

rate as basis functions. The extracted higher-order Chebyshev coefficients (elastic coefficients e_n and viscous coefficients v_n) are all real numbers and possess direct physical significance. For instance, the sign of the third-order elastic coefficient e_3 directly indicates whether the material undergoes strain stiffening ($e_3 > 0$) or strain softening ($e_3 < 0$); the sign of the third-order viscous coefficient v_3 clearly delineates shear thickening ($v_3 > 0$) and shear thinning ($v_3 < 0$) behaviors.

$$e_3 = -|\,G_3^*\,|\cos\,\delta_3 \begin{cases} >0 & \text{for } \pi/2 < \delta_3 < 3\pi/2 \quad \text{strain stiffening} \\ =0 & \text{for } \delta_3 = \pi/2, 3\pi/2 \quad \text{linear elastic} \\ <0 & \text{for } -\pi/2 < \delta_3 < \pi/2 \quad \text{strain softening} \end{cases} \tag{6}$$

$$v_3 = |\,G_3^*\,|/\omega\sin\,\delta_3 \begin{cases} >0 & \text{for } 0 < \delta_3 < \pi \quad \text{shear thickening} \\ =0 & \text{for } \delta_3 = 0, \pi \quad \text{linear viscous} \\ <0 & \text{for } \pi < \delta_3 < 2\pi \quad \text{shear thinning} \end{cases} \tag{7}$$

Compared to the complex representation of Fourier coefficients, the Chebyshev method transforms abstract frequency-domain distortion into concrete physical parameters, allowing the independent quantification of nonlinear elastic and nonlinear viscous characteristics. This intuitive mapping relationship effectively avoids the ambiguity in physical interpretation caused by complex phase angles, providing a clearer mathematical tool to establish the connection between macroscopic rheological responses and microstructural evolution (such as network stretching, chain disentanglement, etc.).

Ewoldt et al. further defined the local elastic responses G'_M and G'_L at small and large instantaneous strains, as well as the instantaneous viscosities (or viscous dissipation coefficients) η'_M and η'_L at minimum and maximum shear rates, and provided their descriptions in terms of both Fourier and Chebyshev coefficients. Furthermore, the S value (strain stiffening ratio) and T value (shear thickening ratio) can be defined.

$$G'_M = \frac{d\sigma}{d\gamma}\,|_{\gamma=0} = \sum_{n=1,odd} n\,G'_n = e_1 - 3e_3 + \cdots \tag{8}$$

$$G'_L = \frac{\sigma}{\gamma}\,|_{\gamma=\pm\gamma_0} = \sum_{n=1,odd}(-1)^{(n-1)/2} G'_n = e_1 + e_3 + \cdots \tag{9}$$

$$S = \frac{G'_L - G'_M}{G'_L} = \frac{4e_3 + \cdots}{e_1 + e_3 + \cdots} \tag{10}$$

$$\eta'_M = \frac{d\sigma}{d\dot{\gamma}}\,|_{\dot{\gamma}=0} = \frac{1}{\omega}\sum_{n=1,odd}(-1)^{(n-1)/2} n G''_n = v_1 - 3v_3 + \cdots \tag{11}$$

$$\eta_L' = \frac{\sigma}{\dot{\gamma}}|_{\dot{\gamma}=\pm\dot{\gamma}_0} = \frac{1}{\omega}\sum_{n=1,odd} G_n'' = v_1 + v_3 + \cdots \quad (12)$$

$$T = \frac{\eta_L' - \eta_M'}{\eta_L'} = \frac{4v_3 + \cdots}{v_1 + v_3 + \cdots} \quad (13)$$

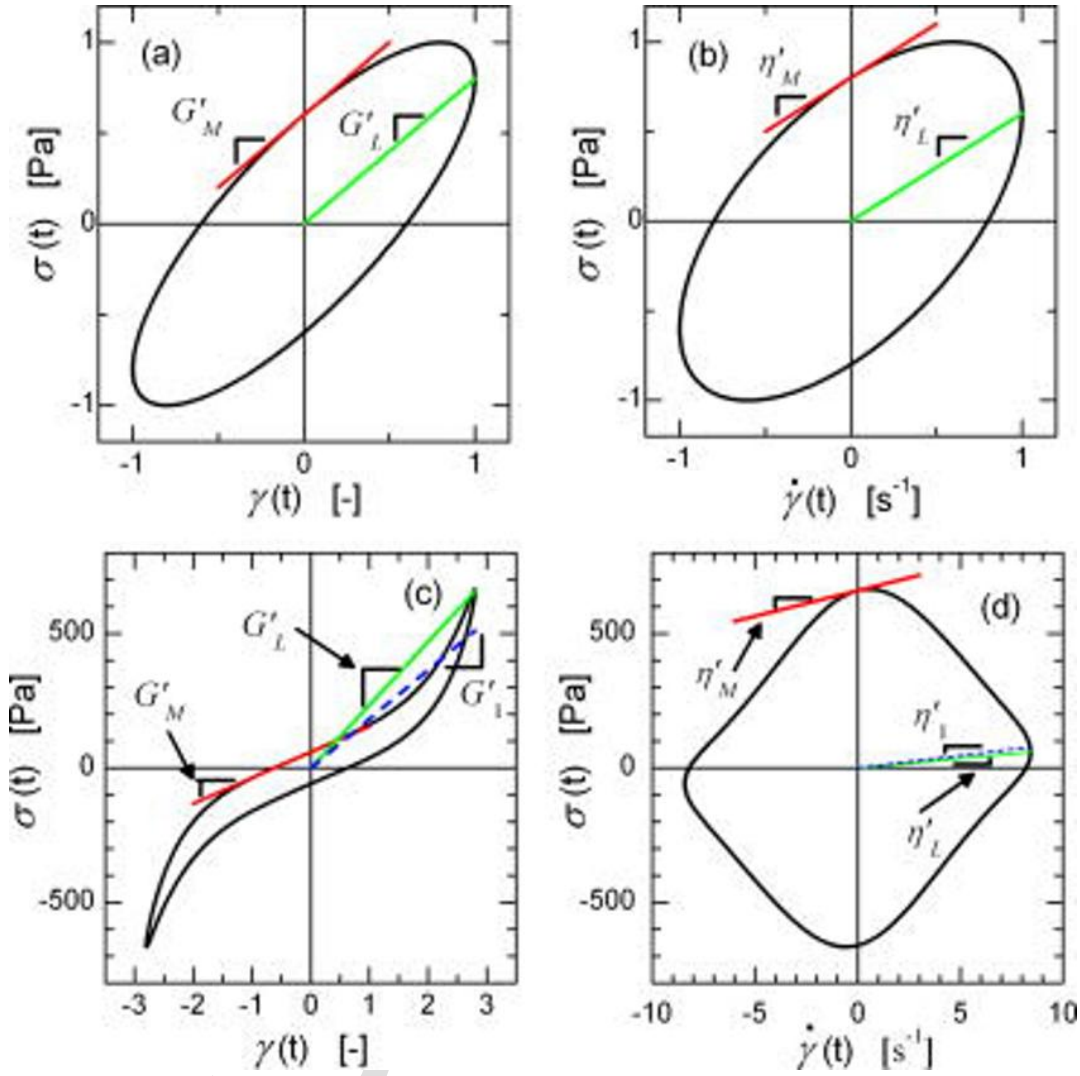


Figure 26. Example Lissajous curves for determining G_M', G_L', η_M', and η_L' under SAOS (a, b) and LAOS (c, d) conditions[10].

4.5.2 Operation Procedure

The operational steps to execute the S-T Values (Chebyshev Polynomials) plotting program are as follows: Enter the main interface of LAOS Analysis, and in the Plot module, click the "S-T Value (Chebyshev Polynomials)" function module under the graph type options.

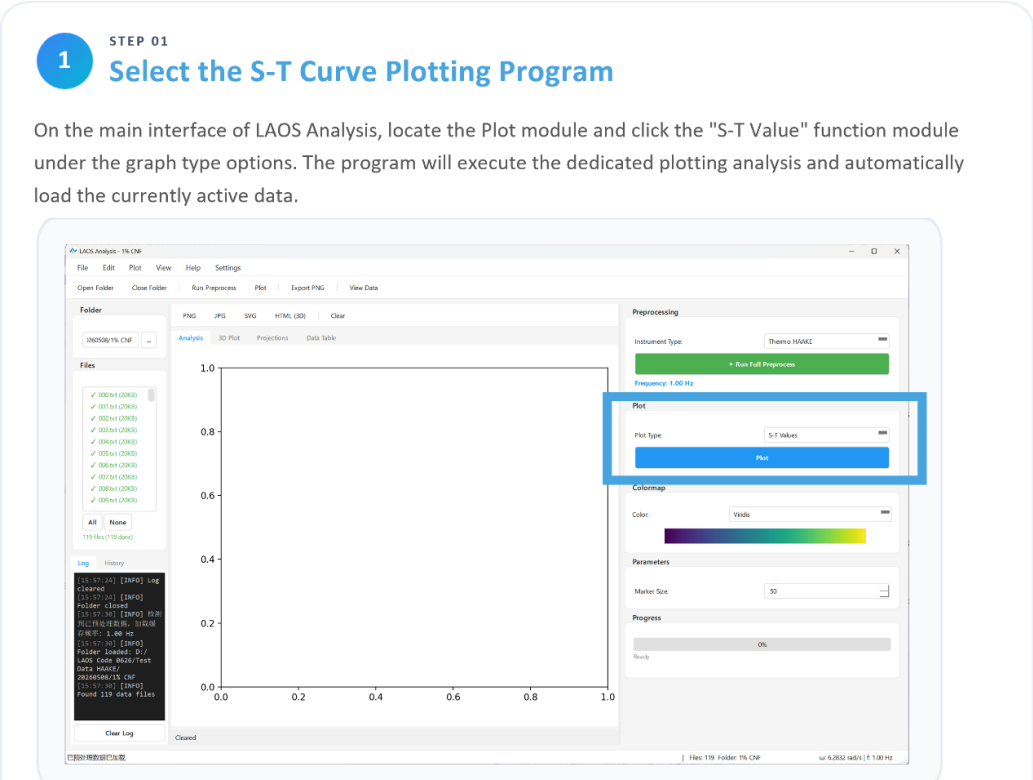


Figure 27. Step settings for the S-T Values (Chebyshev Polynomials) plotting program.

4.5.3 Output Results

Analysis window: Displays the curves of the S value, T value, G_M', G_L', η_M', and η_L' as a function of strain amplitude.

Data Table window: Provides seven columns of data: strain amplitude, S value, T value, G_M', G_L', η_M', and η_L'.

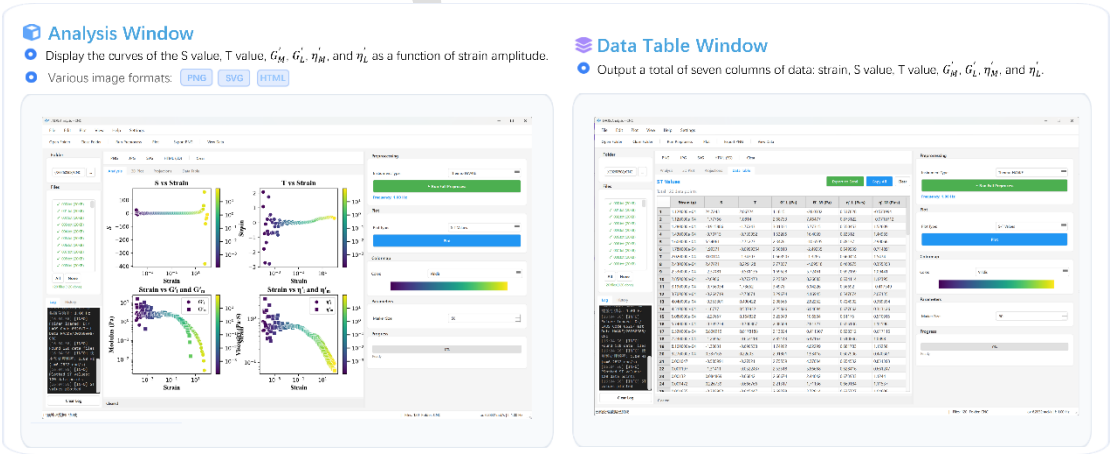


Figure 28. Result display of the S-T Values (Chebyshev Polynomials) plotting program.

4.6 Stress Bifurcation Method

4.6.1 Basic Principles

The definition and measurement of yield stress have long been a contentious issue in rheology, and achieving its scientific, convenient, and accurate characterization remains a challenge to date. Traditional measurement paradigms typically yield a single constant yield stress value, neglecting the interval characteristics of the yielding evolution. To address this, Yu et al. further developed the stress bifurcation method based on geometric averaging. By quantifying the endpoint variations of the geometric mean curves derived from the elastic and viscous Lissajous curves, the stress bifurcation method analyzes the extent to which the elastic Lissajous curve deviates from the ideal linear response, and the viscous Lissajous curve deviates from the ideal circular or elliptical shape. This directly characterizes the breakdown state of the material's microstructure during deformation [11-13].

Calculating the geometric mean curves from the elastic and viscous Lissajous curves and extracting their endpoints yields six specific points: $(\gamma_0, \bar{\sigma}_0)$, $(\bar{\gamma}_0, \sigma_0)$, (γ_0, σ_0), $(\dot{\gamma}_0, \bar{\sigma}_0)$, $(\bar{\dot{\gamma}}_0, \sigma_0)$, $(\dot{\gamma}_0, \sigma_0)$ (Figure 20a, b). By comprehensively analyzing these six points across all strain amplitudes (as shown in Figure 29c), three σ vs. γ curves and three σ vs. $\dot{\gamma}$ curves can be obtained. At low strain amplitudes, the σ vs. γ curves overlap, while the σ vs. $\dot{\gamma}$ curves are separated; conversely, at high strain amplitudes, the σ vs. γ curves separate, while the σ vs. $\dot{\gamma}$ curves overlap. From this, two bifurcation points can be identified. The bifurcation point of the σ vs. γ curves represents the minimum strain/stress required for the stress-strain relationship to deviate from linearity, which is thus termed the onset yield stress. The bifurcation point of the σ vs. $\dot{\gamma}$ curves represents the minimum strain/stress required for the stress-strain rate relationship to approach linearity, which is consequently defined as the end yield stress.

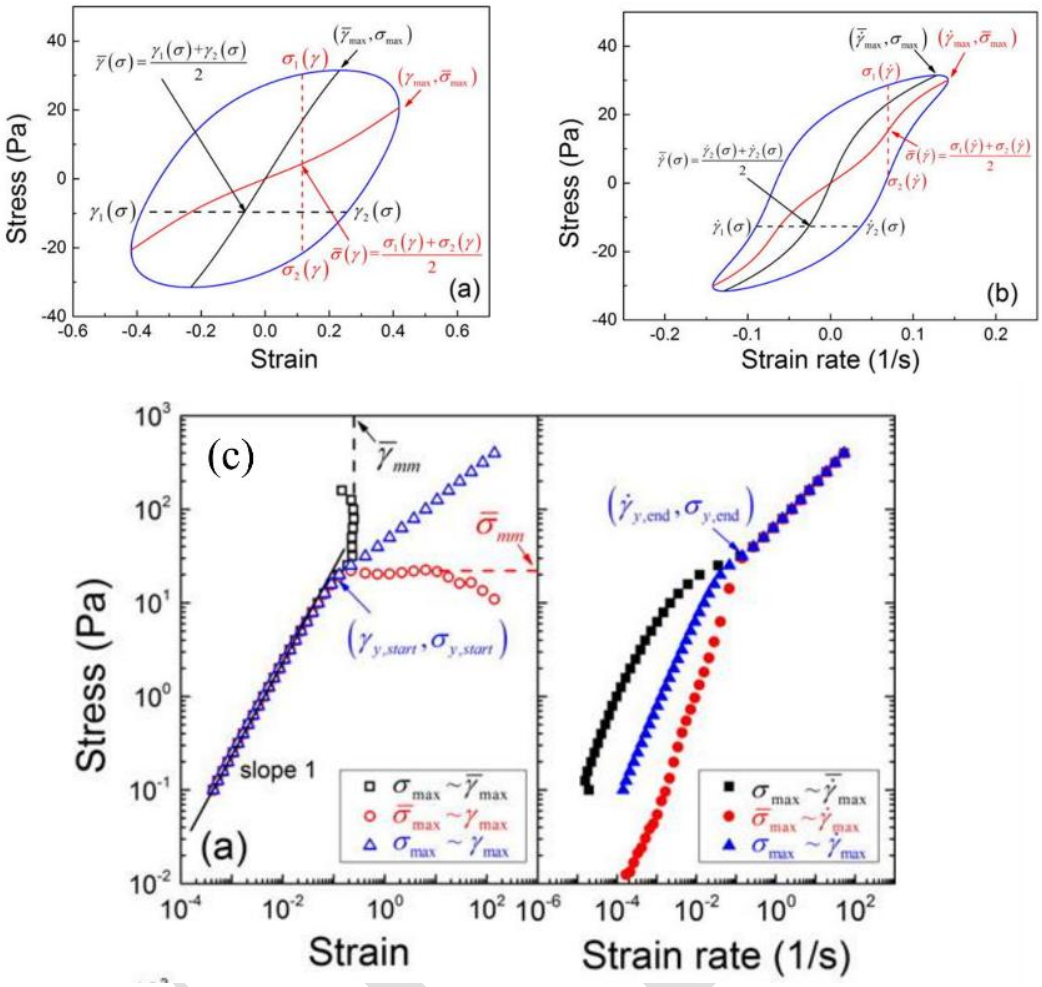


Figure 29. Schematic of the geometric mean for the stress-strain (a) and stress-strain rate (b) Lissajous curves of an 80% PDMS emulsion at 0.05 Hz under a stress amplitude of 31.5 Pa. Schematic of the stress bifurcation among the endpoint curves for the 80% PDMS emulsion under LAOS at 0.05 Hz (c) [11].

4.6.2 Operation Procedure

The operational steps to execute the stress bifurcation plotting program are as follows:

(1) Step 1 Module Selection: Click the "Stress Bifurcation" function module under the graph type options.

(2) Step 2 Parameter Settings: Select whether to enable the automatic determination of bifurcation points (the software identifies the bifurcation point based on whether the relative deviation exceeds 10%).

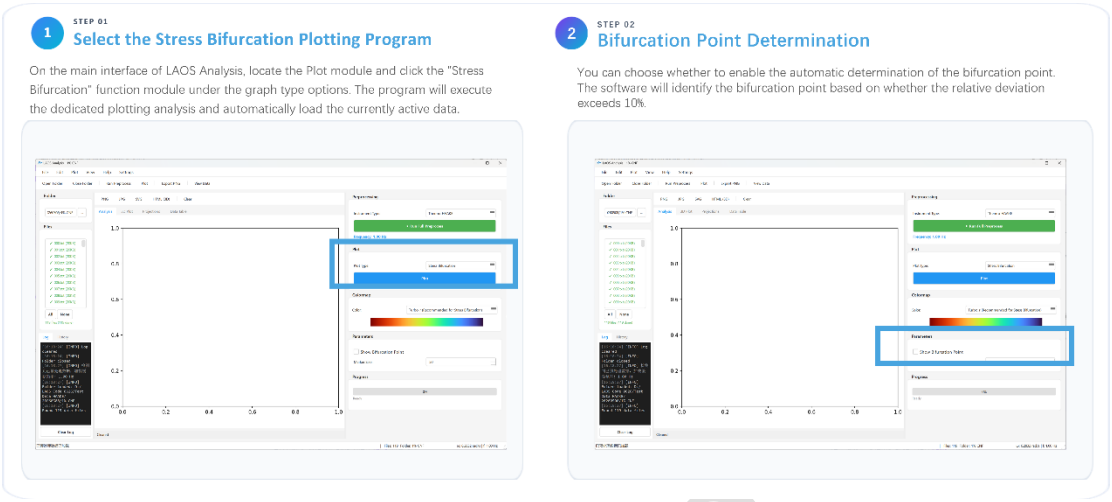


Figure 30. Settings steps for the stress bifurcation plotting program.

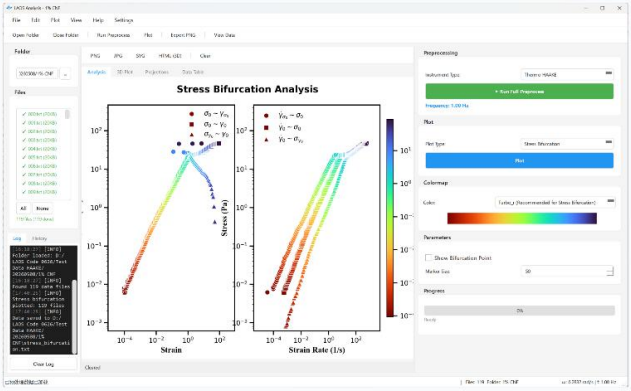
4.6.3 Output Results

(1) Analysis window: Displays the stress-strain bifurcation plot and the stress-strain rate bifurcation plot. Users can toggle the display of the bifurcation points and their corresponding data.

(2) Data Table window: Outputs seven columns of data: σ_0, γ_0, $\dot{\gamma}_0$, σ_{γ_0}, $\sigma_{\dot{\gamma}_0}$, γ_{σ_0}, $\dot{\gamma}_{\sigma_0}$.

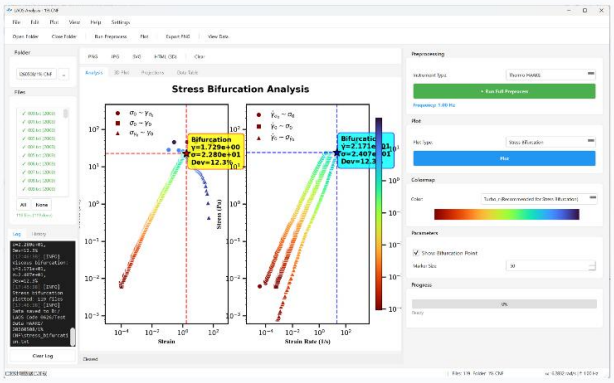
Analysis Window-Show Bifurcation Points Not Selected

- Displays the stress-strain bifurcation plot and the stress-strain rate bifurcation plot without showing the bifurcation points.
- Various image formats: PNG SVG HTML



Analysis Window-Show Bifurcation Points Selected

- Displays the stress-strain bifurcation plot and the stress-strain rate bifurcation plot, showing the bifurcation points and their corresponding data for user reference.



Data Table Window

- Provides seven data columns in total.: $\sigma_0, \gamma_0, \dot{\gamma}_0, \sigma_{\gamma_0}, \sigma_{\dot{\gamma}_0}, \gamma_{\sigma_0}, \dot{\gamma}_{\sigma_0}$

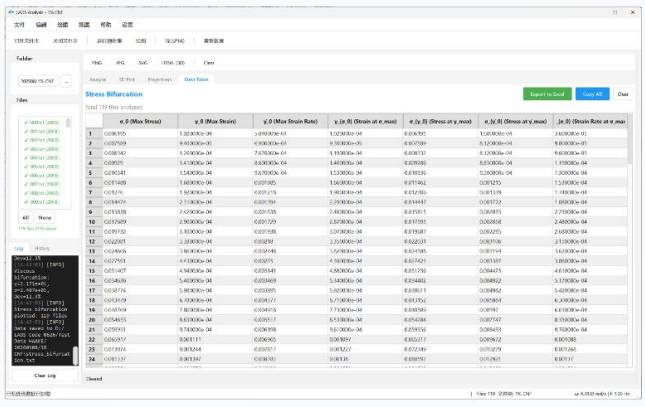


Figure 31. Output results of the stress bifurcation plotting program.

4.7 Sequence of Physical Processes (SPP) Method

4.7.1 Basic Principles

The Sequence of Physical Processes (SPP) is an analytical framework for interpreting the nonlinear response of LAOS. Unlike traditional Fourier transform or Chebyshev polynomial methods, the SPP method no longer treats the entire oscillation cycle as a whole for macroscopic decomposition; instead, it decouples the continuous deformation within a single cycle into a series of sub-processes with clear physical significance. This provides a perspective rich in dynamic detail for gaining insight into the transient evolution of the material's microstructure [14, 15].

Its core principle lies in the fact that during a complete LAOS cycle, complex fluids typically undergo alternating processes: elastic energy storage, microstructural yielding and breakdown, viscous-dominated flow, and thixotropic structural recovery. The SPP method specifically points out that traditional moduli defined by simple ratios or ordinary derivatives cannot truly reflect the local elastic and viscous contributions in the nonlinear regime; therefore, it tends to reconstruct them using partial derivatives. Consequently, SPP defines two new characteristic parameters: the instantaneous elastic modulus G_t' and the instantaneous viscous modulus G_t''. By tracking the evolution trajectories of these two instantaneous moduli, or by plotting the state trajectory in the three parametric space of stress-strain-strain rate ($\sigma - \gamma - \dot{\gamma}$), SPP can accurately characterize the dominant physical mechanism of the material at each phase point.

To further quantify this dynamic evolution, the SPP method introduces the time derivatives of the instantaneous moduli, $dG_t'(t)/dt$ and $dG_t''(t)/dt$, to describe the direction and rate of change in the $G_t'(t)$ and $G_t''(t)$ trajectories. The physical significance of the signs of these derivatives is highly intuitive: $dG_t'(t)/dt > 0$ implies an enhancement of the instantaneous elastic response, characterizing strain hardening; $dG_t'(t)/dt < 0$ characterizes strain softening; similarly, $dG_t''(t)/dt > 0$ indicates an increase in instantaneous viscous resistance, i.e., shear thickening; while $dG_t''(t)/dt < 0$ corresponds to shear thinning.

$$\frac{d\sigma(\gamma,\dot{\gamma})}{d\gamma} = \frac{\partial\sigma}{\partial\gamma} + \left(\frac{\partial\sigma}{\partial\dot{\gamma}}\right)\left(\frac{\partial\dot{\gamma}}{\partial\gamma}\right), G_t' \equiv \frac{\partial\sigma}{\partial\gamma} \tag{14}$$

$$\frac{d\sigma(\gamma,\dot{\gamma})}{d\dot{\gamma}} = \frac{\partial\sigma}{\partial\dot{\gamma}} + \left(\frac{\partial\sigma}{\partial\gamma}\right)\left(\frac{\partial\gamma}{\partial\dot{\gamma}}\right), G_t'' \equiv \frac{\partial\sigma}{\partial(\dot{\gamma}/\omega)} \tag{15}$$

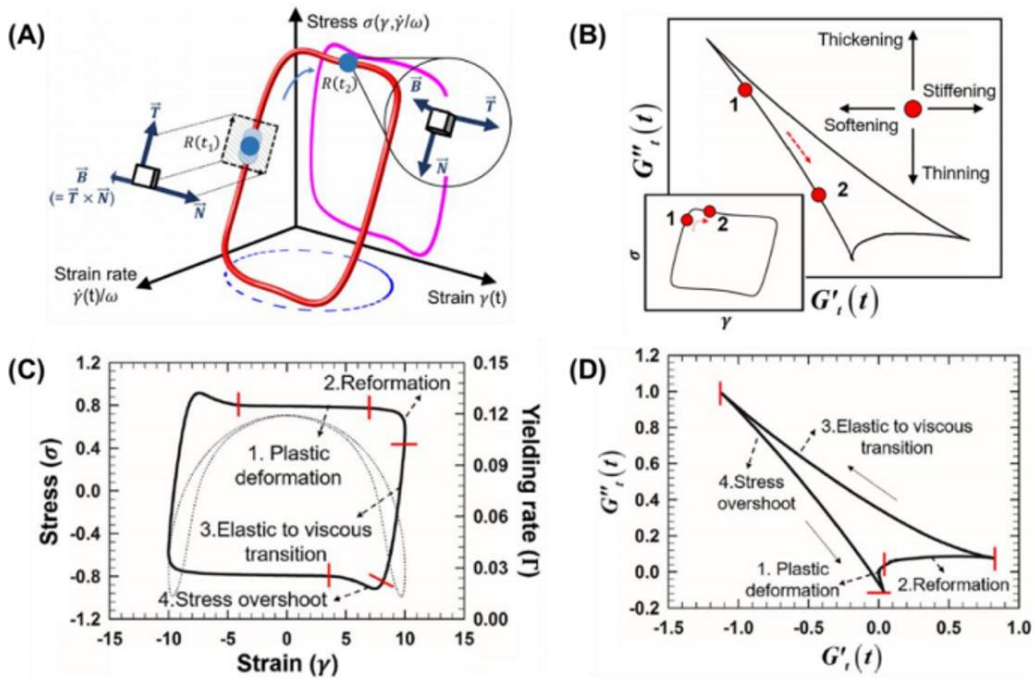


Figure 32. (A) An example of the stress response under LAOS as a 3D spatial trajectory. (B) Description of rheological transitions in the Cole-Cole plot within the SPP method. (C) Lissajous curve of the steady-state LAOS response ($\gamma_0 = 10$). Four processes are identified: plastic deformation, structural reconstruction, viscoelastic transition, and stress overshoot. The butterfly-shaped dashed line indicates the yielding rate. (D) Cole-Cole plot of G_t' and G_t'' under the same steady-state conditions [15].

4.7.2 Operation Procedure

4.7.2.1 Single SPP Plotting

This refers to plotting the G_t'-G_t'' and $dG_t'(t)/dt$-$dG_t''(t)/dt$ corresponding to a specific strain amplitude. The operation procedure to execute the single SPP plotting program is as follows:

(1) Step 1 Module Selection: Enter the main interface of LAOS Analysis, click and select the "SPP Method" function module in the graph type options of the Plot module, and further specify the "Calculate & Plot" mode.

(2) Step 2 Data Loading: In the file index field, enter the target txt data file for the single plot analysis.

(3) Step 3 Parameter Settings: In the parameter configuration area, set the number of Fourier orders involved in stress reconstruction according to analysis requirements

to control fitting accuracy; the system default is set to 3rd order.

(4) Step 4 Range Settings: Define the data range. Due to symmetry, a single strain oscillation cycle can yield two overlapping SPP curves; users can choose to use either the full data or half of the data for plotting.

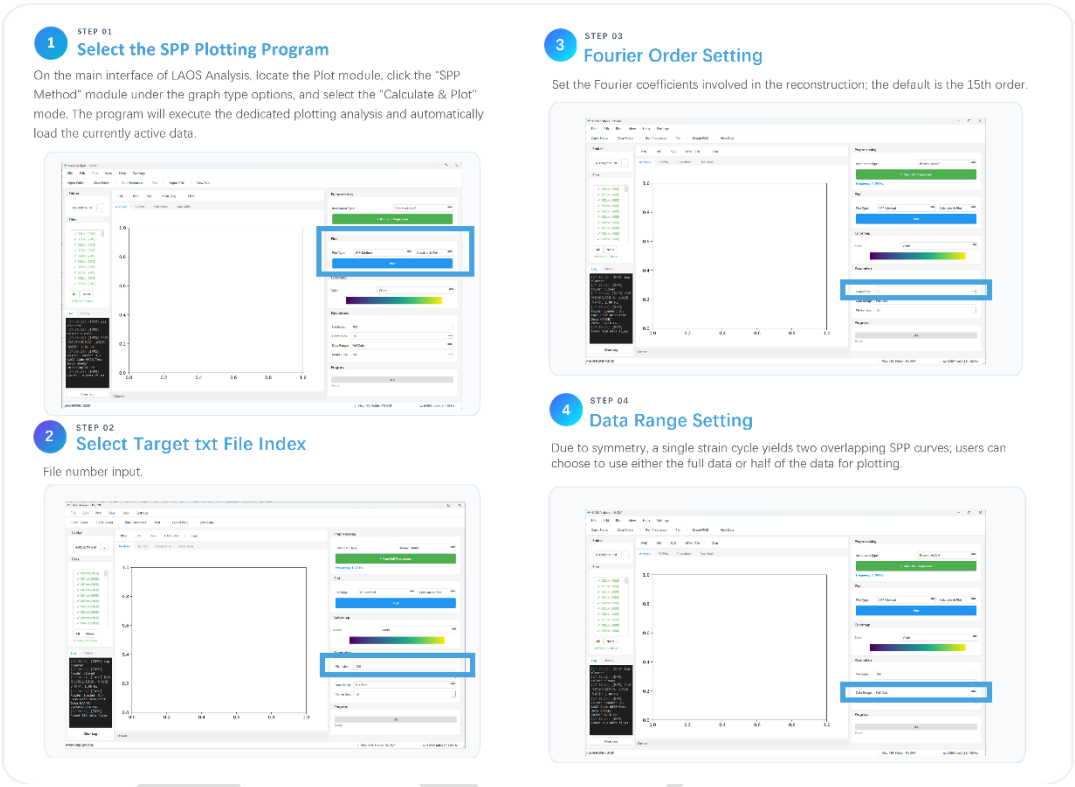


Figure 33. Step settings for the single SPP plotting program.

4.7.2.2 Multi-File SPP Overlay

This refers to simultaneously plotting the $G_t' - G_t''$ and $dG_t'/dt - dG_t''(t)/dt$ corresponding to multiple specified strain amplitudes. The operation procedure to execute the multi-file SPP overlay plotting program is as follows:

(1) Step 1 Module Selection: In the graph type options of the Plot module, click and select the "SPP Method" function module, and switch to the "Multi-File Overlay" mode to achieve comparative analysis of multiple datasets.

(2) Step 2 Data Loading: In the multi-file number input box, type the corresponding numbers of the files to be overlaid. Note that the numbers must be strictly separated by English commas.

(3) Step 3 Parameter Settings: In the parameter configuration area, set the Fourier order for the multi-file overlay calculation; the system default value is 3rd order.

(4) Step 4 Range Settings: Define the data range. Due to symmetry, a single strain oscillation cycle can yield two overlapping SPP curves; users can choose to use either the full data or half of the data for plotting.

(5) Step 5 Label Settings: Enter a custom curve title for each file number; these titles will be displayed in the legend.



Figure 34. Step settings for the multi-file SPP overlay plotting program.

4.7.3 Output Results

4.7.3.1 Single SPP Plotting

(1) Analysis window: Displays the Cole-Cole plots of G_t'-G_t'' and dG_t'/dt-$dG_t''(t)/dt$, intuitively presenting the dynamic mechanical evolution trajectory of the material under a single condition.

(2) Data Table window: Outputs four columns of core data, namely G_t', G_t'',

dG_t'/dt, and dG_t''/dt, recording in detail the instantaneous moduli and rate of change characteristics during the SPP evolution process.

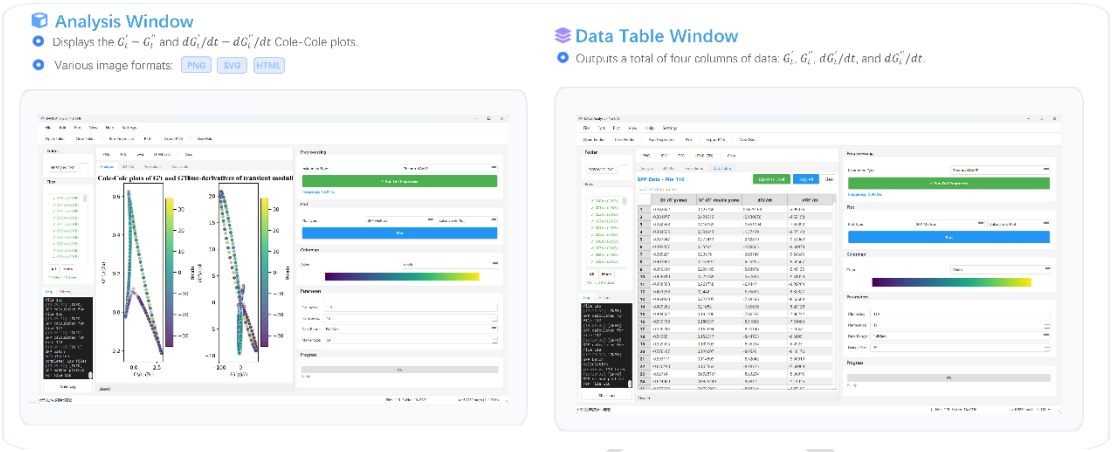


Figure 35. Output results of the single SPP plotting program.

4.7.3.2 Multi-File SPP Overlay

(1) Analysis window: Overlays and displays the Cole-Cole plots of G_t'-G_t'' and dG_t'/dt - $dG_t''(t)/dt$ under different strain conditions, facilitating an intuitive comparison of the SPP trajectory evolution characteristics at different strain levels.

(2) Data File: Generates the reconstructed stress, strain, and strain rate data files along with the SPP data files in the specified file path, for subsequent in-depth analysis and archiving.

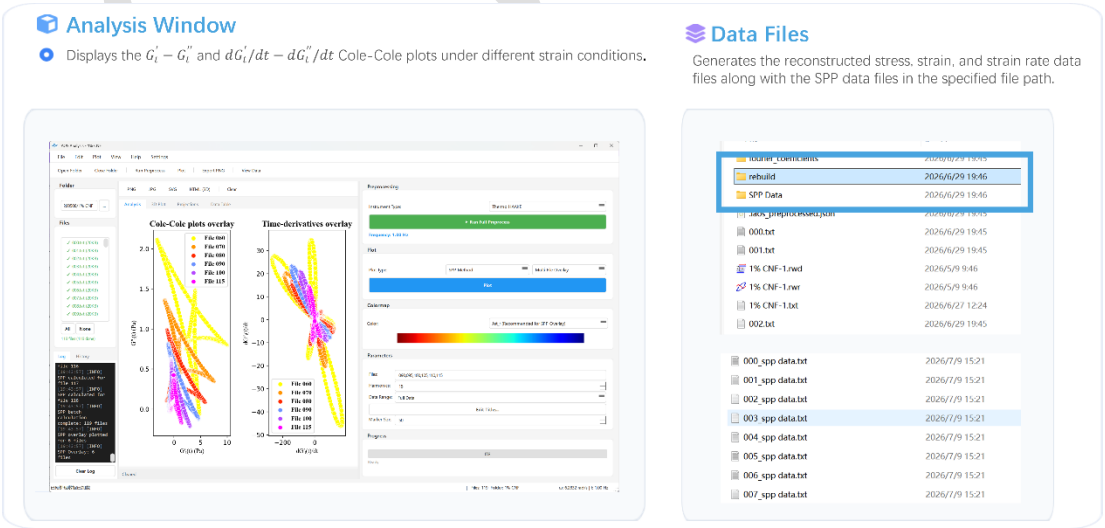


Figure 36. Output results of the multi-file SPP overlay plotting program.

4.8 Cage Modulus

4.8.1 Basic Principles

In dense soft matter systems, due to extremely high particle concentrations or severe spatial confinement, any given central particle is surrounded by closely packed neighboring particles, forming a transient local space referred to as a "cage". In oscillatory shear experiments, the cage undergoes cyclic elastic deformation, yielding, flow, and reformation at the strain extremes. The cage modulus is defined as the instantaneous slope at the zero-stress point on the elastic Lissajous curve, which quantifies the resistance to deformation after each reformation of the cage [16].

$$G_{cage} = \frac{d\sigma}{d\gamma}\,|_{\sigma=0} \tag{16}$$

For oscillatory testing:

$$\gamma = \gamma_0 \sin\ \omega t \tag{17}$$

$$\sigma(t, \omega, \gamma_0) = \gamma_0 \sum\nolimits_{n=\text{odd}} (G_n' \sin(n\omega t) + G_n'' \cos(n\omega t)) \tag{18}$$

$$G_{cage} \equiv \frac{d\sigma}{d\gamma}\,|_{\sigma=0} = \frac{1}{\cos(\omega t)} \sum\nolimits_{n=\text{odd}} \{nG_n' \cos(n\omega t) - nG_n'' \sin(n\omega t)\}\,|_{\sigma=0} \tag{19}$$

For the elastic limit case, i.e., $n = 1$ and $G_n'' \to 0$, the cage modulus is:

$$\lim_{n\to 1, G_n''\to 0} \frac{d\sigma}{d\gamma}\,|_{\sigma=0} = \frac{1}{\cos(\omega t)} \{G' \cos(\omega t)\} = G' \tag{20}$$

For linear viscoelasticity, i.e., $n = 1$, the cage modulus is:

$$G_{cage} = G' + \frac{(G'')^2}{G'} \tag{21}$$

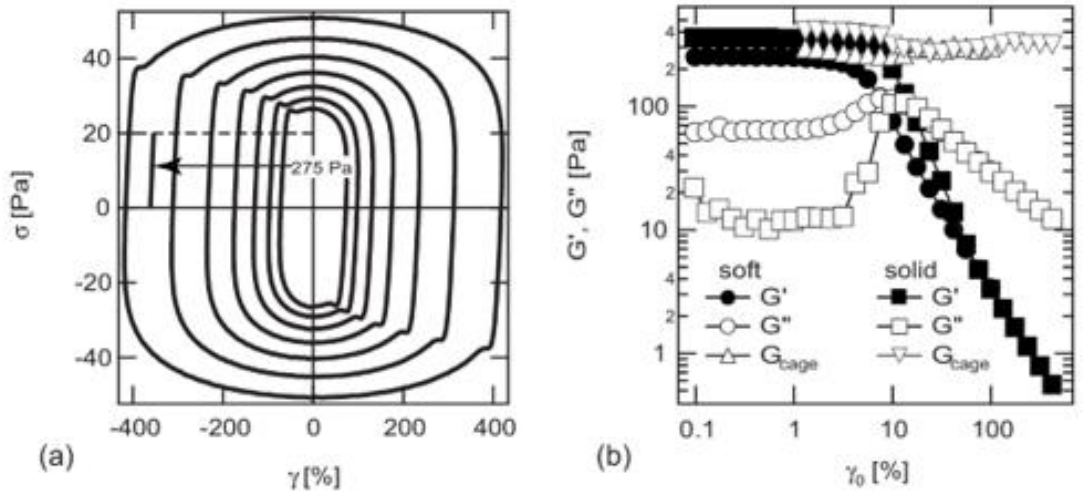


Figure 37. (a) Lissajous curves of a multi-arm polybutadiene solution under oscillatory tests in the solid-state experiment (more than 4000 s after pre-shear, sample aging occurs); (b) Elastic modulus, viscous modulus, and apparent cage modulus in the soft-state experiment (approximately 100–1000 s after pre-shear, incomplete structural recovery) and solid-state experiment [16].

4.8.2 Operating Procedure

The operating procedure for the cage modulus plotting program is as follows:

(1) Step 1 Module Invocation: Enter the Plot module, and in the graph type options, click and enable the "Cage Modulus" function module to evaluate and analyze the elastic stiffness of the cage structure and the degree of linear elastic recovery.

(2) Step 2 Data Selection: Based on actual analysis requirements, one can choose to plot the "G_{cage} vs Strain" curve to observe the evolution of the cage modulus, or choose to plot the "G'/G_{cage} vs Strain" curve to measure the degree of linear elastic recovery of the material.

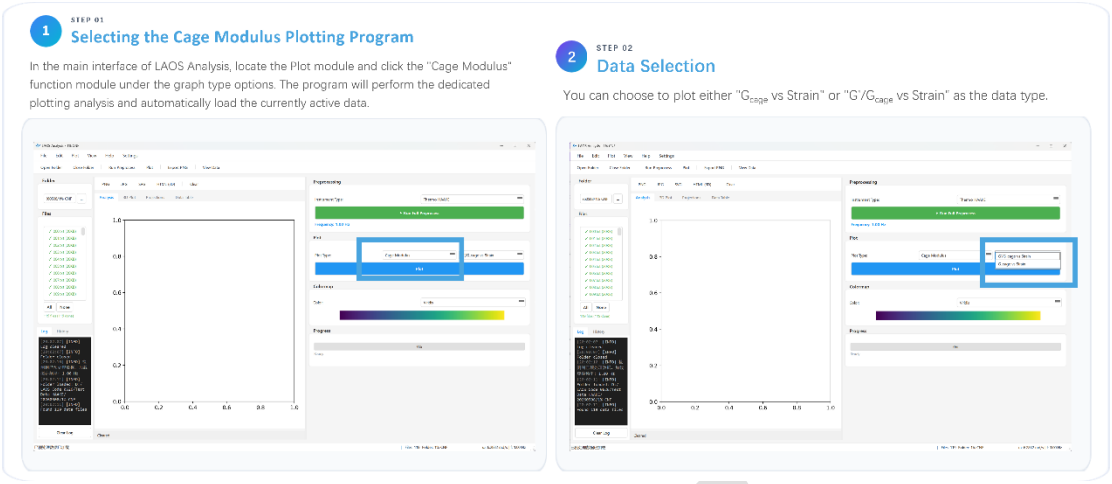


Figure 38. Step operation settings for the cage modulus plotting program.

4.8.3 Output Results

(1) Analysis Window: Intuitively renders and displays the characteristic curves of G_{cage} or G'/G_{cage} varying with strain, clearly presenting the strain-dependent evolution of the cage structure's elastic stiffness or the material's linear elastic recovery capability.

(2) Data Table Window: Accurately outputs two columns of core data, namely strain and its corresponding calculated values of G_{cage} or G'/G_{cage}, providing reliable support for subsequent quantitative analysis and data comparison.

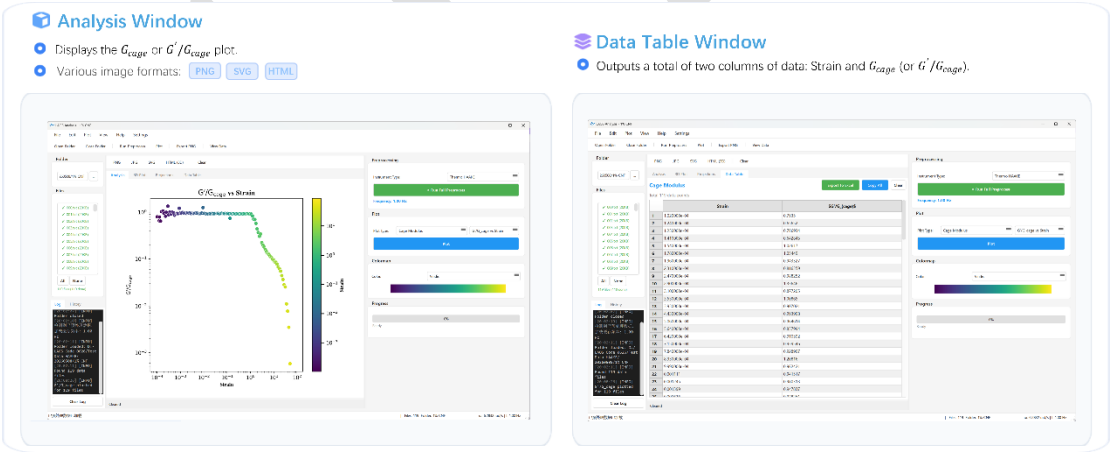


Figure 39. Output results of the cage modulus plotting program.

Chapter 5 Common Errors and Troubleshooting

5.1 Program Startup Issues

Issue Description: The program fails to start or immediately crashes after double-clicking LAOS_Analysis.exe.

Example Error Message: ImportError: DLL load failed while importing numpy

Solution: If you encounter this issue, it is recommended to first check whether the necessary runtime environments are installed on the system, and confirm that the file path does not contain spaces or Chinese characters. Also, ensure the program folder is complete, especially the inclusion of the _internal directory. If the problem persists, try running the program as an administrator. Additionally, check if antivirus software is blocking the program and add the program directory to the trust whitelist if necessary.

5.2 Language Switching Issues

Issue Description: After switching languages via Settings→Language, the interface still displays in English.

Example Error Message: [DEBUG LANG] Warning: Translation file not found: zh_CN.ts

Solution: Typically, the interface will automatically refresh after switching languages; if it does not take effect, try restarting the program. Meanwhile, ensure that the _internal\translations directory and contains the zh_CN.ts and ja.ts language pack files. Note that when the language option is set to Auto, the system will automatically detect and match the operating system's language settings.

5.3 Data Loading Issues

5.3.1 Unable to Open Folder

Issue Description: After selecting a folder via File→Open Folder, the file list appears empty.

Example Error Message: [DEBUG] No .txt files found in selected folder

Solution: Please ensure that the selected folder contains data files in the correct format (e.g., numbered files like 0.txt, 1.txt, etc.). Secondly, check that the file encoding is UTF-8 or ANSI. Most importantly, the file content must conform to the specification, containing four columns of data: time, stress, strain, and strain rate; otherwise, it cannot

be recognized properly.

5.3.2 Data Format Error

Issue Description: Loaded data displays abnormally or triggers an error.

Example Error Message: ValueError: could not convert string to float: Time

Solution: This error is usually caused by the data file containing a header row; it is recommended to delete the first row (header). Meanwhile, ensure that data columns are separated by spaces, commas, or tabs, and check the file for blank lines or invalid data.

5.4 Plotting Issues

5.4.1 Plot Shows No Data

Issue Description: After clicking the Plot button, the plotting area is blank.

Example Error Message: IndexError: list index out of range - file_index: 150

Solution: First, ensure that data preprocessing (Run Full Preprocess) has been completed. Secondly, check if the File Index is within the valid range (0 to total number of files minus 1), and confirm that the Plot Type is selected correctly. If the issue remains unresolved, check the error messages in the bottom Log Panel to pinpoint the specific cause.

5.4.2 Menu Plotting Fails to Execute

Issue Description: After selecting a plot type from the menu and confirming in the parameter dialog box, no plot is generated.

Example Error Message: KeyError: plot_type not found in params

Solution: Ensure that the data folder has been successfully opened, and accurately fill in the required parameters in the parameter dialog box: for example, File Index is required for Lissajous curves, Files (e.g., 030,120,150) for Lissajous Superposition, and Harmonics for Fourier coefficients. After completing the parameter inputs and clicking OK to confirm, the system should automatically execute the plotting.

5.4.3 3D Plot Export Failure

Issue Description: Exporting the 3D Lissajous plot to PNG/JPG/SVG format fails.

Example Error Message: Export Format Not Supported: 3D plot image export

(PNG/JPG/SVG) is not supported

Solution: It is recommended to export 3D images in HTML format instead. The HTML format not only has a higher export success rate but also preserves interactive 3D visualization features. Please note that the current packaged version of the program does not support static format export for 3D images.

5.4.4 Incomplete Higher-Order Fourier Coefficient Data Table

Issue Description: The higher-order Fourier coefficient data table is displayed incompletely or fails to export.

Example Error Message: TypeError: int() argument must be a string or number, not NoneType

Solution: You need to control the display order by setting the Max Harmonic (n) parameter (the system supports a maximum of 50 orders). If exporting to Excel fails, ensure that the selected save path has proper write permissions.

5.5 Export Issues

5.5.1 Poor Image Export Quality

Issue Description: Exported images have low resolution or display font abnormalities.

Example Error Message: UserWarning: DPI set too low for print quality (current: 72)

Solution: You can improve clarity by setting an appropriate DPI (150-600 recommended) via Settings→Export Settings, and adjust the font size and type in Settings→Plot Font to improve the display. For format selection, PNG is recommended for the best image quality.

5.5.2 No Permission for Export Path

Issue Description: An error occurs when exporting files, unable to save.

Example Error Message: PermissionError: [Errno 13] Permission denied: C:\Windows\output.png

Solution: This issue is mostly caused by the lack of write permissions for the target path. Do not save directly to system directories like C:\Windows. It is recommended to choose paths with write permissions, such as the Desktop or Documents, and ensure

that the target disk has sufficient storage space.

5.6 Preprocessing Issues

5.6.1 Preprocessing Frequency Input Error

Issue Description: The frequency parameter is set incorrectly during preprocessing.

Example Error Message: Frequency Input Error: omega = 0, please enter valid frequency

Solution: Clarify the selection logic for frequency units: if Hz (standard frequency unit) is selected, entering 1 means 1 Hz; if rad/s (angular frequency) is selected, entering 2 means 2π rad/s (i.e., 1 Hz). Make sure the frequency value entered in the software is completely consistent with the actual experimental setup parameters.

5.6.2 Preprocessing Progress Stuck

Issue Description: The preprocessing progress bar stops moving or is unresponsive for a long time.

Example Error Message: [DEBUG] Processing timeout: file 150 of 200

Solution: It is normal for processing to take a long time when dealing with a large number of data files; you can check the Log Panel for real-time status. If it remains completely frozen and unresponsive for a long time, it is recommended to restart the program and try again.

5.7 Interface Display Issues

Issue Description: The file list, log panel, or parameter panel is not displayed.

Example Error Message: AttributeError: Control Panel widget not found

Solution: This is mostly caused by panels being manually hidden. You can check the display status of each panel through the View→Show/Hide menu. Users can individually control the visibility of the File & Log Panel, Plot Panel, and Control Panel from here.

5.8 Plot Style Issues

Issue Description: Curve colors, line widths, and other styles do not meet requirements.

Solution: Personalized adjustments can be made in Settings→Plot Style, including Line Width, Marker Size, Marker Shape, and Transparency. Additionally, you can select an appropriate Colormap in Settings→Colors to change the color mapping scheme.

5.9 Slow Program Performance

Issue Description: The program responds slowly or plotting stutters/lags.

Example Error Message: MemoryWarning: High memory usage detected (>2GB)

Solution: It is recommended to reduce the number of simultaneously opened data files, or appropriately reduce plotting parameters (such as decreasing Max Harmonic) to alleviate the computational load. Additionally, closing temporarily unnecessary panels can effectively reduce memory usage, thereby improving the smoothness of the program's operation.

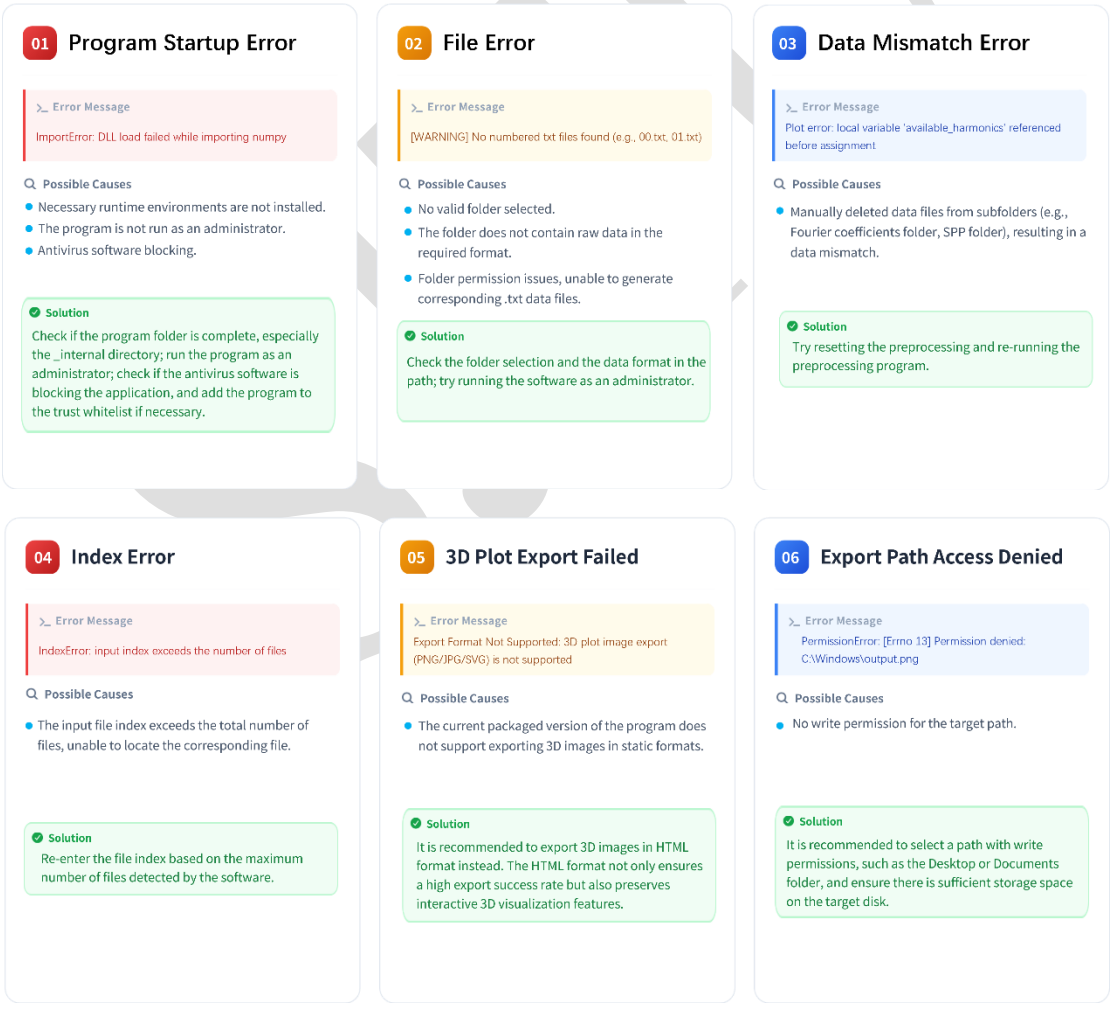


Figure 40. Common errors and solutions.

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